

Networked Competitive Bivirus SIS spread with Higher Order Interactions [☆]

Sebin Gracy^a, Brian D.O. Anderson^b, Mengbin Ye^c, César A Uribe^d

^aDepartment of EECS, South Dakota School of Mines and Technology, Rapid City, SD, USA, (sebin.gracy@sdsmt.edu).

^bSchool of Engineering, Australian National University, Canberra, Australia, (brian.anderson@anu.edu.au).

^cCentre for Optimisation and Decision Science, Curtin University, Australia, (mengbin.ye@curtin.edu.au).

^dDepartment of Electrical and Computer Engineering, Rice University, Houston, TX, USA, (cauribe@rice.edu).

Abstract

The paper studies the simultaneous spread of two competing viruses over a network of population nodes with higher-order interactions (HOI) using a continuous-time time-invariant competitive bivirus networked susceptible-infected-susceptible (SIS) system. In this paper, by HOI, we mean interactions among group sizes of no more than three nodes. The first key contribution is to establish several important general properties for generic systems. Namely, there are a finite number of equilibria, each equilibrium is nondegenerate, and the system is a strongly monotone dynamical system. Put together, we establish that for almost all initial conditions, the system will converge to a stable equilibrium (of which there may be many). We then turn our focus to characterizing the existence and stability of the equilibria of this system, which are i) the disease-free equilibrium (DFE), ii) single-virus endemic equilibria, and iii) coexistence equilibria (where both viruses are present). We present a range of conditions on the existence or nonexistence of various equilibria. Two key features underpin our results: First, we substantially relax the connectivity conditions of the network relative to existing literature. More specifically, for securing several important general properties for generic systems, we do *not* require strong connectivity of the pairwise interaction graph. Second, we identify dynamical phenomena, including multiple stable equilibria, which are known to be impossible without HOI. The latter illustrates the novel insights that are obtained by including HOI into models of epidemic spread. Finally, we illustrate our results using several numerical examples.

Keywords: Epidemic processes, competing viruses, finiteness of equilibria, coexistence equilibrium

1. Introduction

Mathematical modeling of epidemic spreading processes has garnered the interest of multiple scientific communities, and has proven to be critical to policymakers and governments in planning responses to epidemics as highlighted by the COVID-19 pandemic (Giordano et al., 2020; Ferguson et al., 2020). Compartmental models are a fundamental paradigm, where individuals transition between mutually exclusive health compartments. Popular implementations include the susceptible-infected-recovered (SIR), the susceptible-infected-susceptible (SIS) (Brauer et al., 2008; Pastor-Satorras et al., 2015), and the susceptible-asymptomatic-infected-recovered (SAIR) models (Stella et al., 2022).

The present paper deals with a networked SIS model, first studied by (Lajmanovich and Yorke, 1976). In this model, each node in the network represents a large population, and the interconnection among the nodes indicating

the pathways for the virus to spread. Initially, networked SIS models considered a *single* virus circulating in the network. In the past decade, there has been growing interest in multiple viruses *simultaneously* circulating in the network. In such settings, the viruses can be competitive (Liu et al., 2019) or cooperative (Gracy et al., 2025). In the competitive case, as considered in this paper, infection with one virus rules out the possibility of *simultaneous* infection with another virus. Examples include the simultaneous circulation of multiple strains of influenza A virus (Nuno et al., 2005); the spread of different strains of gonorrhea (Castillo-Chavez et al., 1996); and the spread of influenza A virus and rhinovirus (Nickbakhsh et al., 2019). Outside of epidemics, the notion of competitive viruses can be used to understand product adoption in a marketplace (Aral and Walker, 2011), the spread of opinions in social networks (Myers and Leskovec, 2012), individuals' opinions about politicians from opposing parties (Trpevski et al., 2010), and in alternative farming practices (Paré et al., 2021).

The dynamics of competitive networked multi-virus SIS models are more involved than their single-virus counterparts (Ye and Anderson, 2022). Specifically, the complexity stems from the fact that, instead of being concerned

[☆]Part of this material is based upon work supported by the National Science Foundation under Grants #2211815, #2213568, and #2443064. The work of MY is funded by the Western Australian Government through the Premier's Science Fellowship Program, and by the Australian Government through the Office of National Intelligence (NI240100203).

with merely the survival or extinction of a single virus, one must consider the survival or extinction of multiple viruses, including simultaneous survival (coexistence). A key distinguishing property of multi-virus models is that of multistability, i.e., a system has multiple locally stable equilibria and the equilibrium point to which the system converges depends on the initial condition. However, classical multi-virus SIS models do not account for the impact of higher-order interactions (HOIs) in the transmission of the virus.

Broadly speaking, HOIs are interactions whereby the effect is present only when three or more individuals in a population node or possibly across different population nodes simultaneously interact, in contrast to the pairwise interactions of classical networks (Battiston et al., 2020). There is a growing interest in modeling HOIs in networks (not only for epidemics but also in social and ecological contexts). In particular, HOIs have been linked to phenomena and dynamical regimes not possible with just pairwise interactions (Iacopini et al., 2019; Grilli et al., 2017; Mayfield and Stouffer, 2017; Gibbs et al., 2022). This claim has been further strengthened by contributions pertaining to the spread of opinions on Facebook; see (Ugander et al., 2012). Other examples include socio-economic systems where activity is often jointly coordinated between multiple agents (e.g., buyer, seller, broker) (Bonacich et al., 2004); and biochemical systems where reactions may often involve more than two species, or two reagents might interact only in the presence of an enzyme (Klamt et al., 2009). In the context of epidemics, super-spreader events that involve large groups of people interacting in a short time window are of particular concern (Lemieux et al., 2021), and, since classical epidemic models are based on dynamical pairwise interactions between individuals, and hence cannot capture and reproduce such large group interactions, HOIs have been recently proposed to capture such super-spreader events (Wang et al., 2024; Zino and Rizzo, 2024).

The present paper focuses on the spread of two competing viruses over a networked SIS system, with the infections captured by a (possibly) directed hypergraph. In our context, HOIs capture a susceptible individual interacting *simultaneously* with more than one infected individual, but no more than two infected individuals. Hypergraphs, viz., graphs in which an edge can connect more than two nodes, provide one tool to model HOIs, with simplicial networks being a different method (Battiston et al., 2021). SIS networks with HOIs, as captured by a hypergraph, have been studied previously. An SIS model on a hypergraph, albeit with certain restrictions imposed on the network structure, has been studied by (de Arruda et al., 2020), building on the approach of Iacopini et al. (2019), which is a scalar model. The findings by (Iacopini et al., 2019) have been extended to the network case by (Cisneros-Velarde and Bullo, 2021). These aforementioned works only consider a single virus, as opposed to multiple competing viruses. This limitation has been partly overcome by (Cui et al.,

2025) with a bivirus competitive networked SIS model with HOIs (i.e., two competing viruses).

Paper Contributions

The present paper analyzes a competitive networked bivirus SIS model with HOIs. Our contributions significantly extend the existing literature on such models via two core streams.

First, we establish general properties of generic systems to provide insight and global understanding of the dynamics of competitive virus spread on networks with HOIs. The term generic system connotes generic parameter choices; in contrast, nongeneric systems have parameters that can be characterised by an algebraic or semi-algebraic set. We prove that the system is a strongly monotone dynamical system (Theorem 1). The Parametric Transversality Theorem from differential topology is used to establish that for generic systems, there are a finite number of equilibria and each equilibrium is nondegenerate, viz., the Jacobian is nonsingular at the equilibrium (Theorem 2). These two properties allow us to establish a third important conclusion on the limiting behavior of the system (Theorem 3): for almost all initial conditions except possibly a set of measure zero, the system converges to a stable equilibrium. This allows us to set aside many questions of asymptotic behavior, and focus on characterization of equilibria. A key feature of our findings in Theorems 1-3 is that we do *not* require strong connectivity of the pairwise interaction digraph - this is in sharp contrast with (Cui et al., 2025, Theorem 10).

As noted in the abstract, there are three types of equilibria, viz., disease-free equilibrium (DFE), single-virus endemic equilibrium, and coexistence equilibria. Our second stream identifies a number of different conditions (both necessary, and separately, sufficient) that characterize the existence (or nonexistence) and local stability properties of the equilibria. This includes the existence and local stability of the DFE and single-virus endemic equilibria corresponding to each of the two viruses (Proposition 2), results on the system when only one virus is present (Theorem 4) and conditions for the existence or nonexistence of coexistence equilibria (Theorem 5, and Proposition 6, respectively). Proposition 2 highlights an interesting phenomenon. Namely, we identify a class of parameters that permits tristability (three locally stable equilibria), the first being the DFE and the other two being single-virus endemic equilibria corresponding to each of the two viruses. The phenomenon of simultaneous stability of multiple endemic equilibria has been observed in the context of spread of two strains of tuberculosis, a drug-sensitive one and a multi-drug-resistant strain (Gumel and Song, 2008). Note that in the classical bivirus SIS setting, it is impossible for the DFE and an endemic equilibrium to be simultaneously stable (Ye and Anderson, 2022, Section E), showcasing the impact of HOIs.

As a byproduct, we also make the following auxiliary contributions. For the directed hypergraph associated with

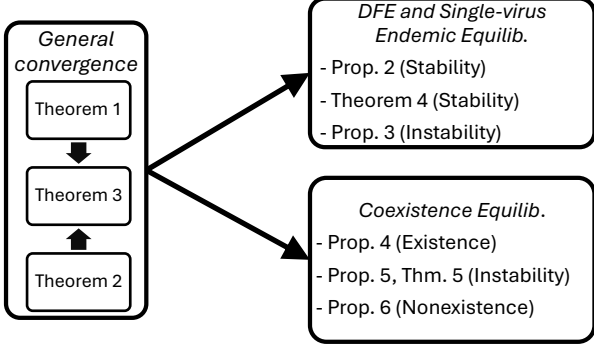


Figure 1: A summary of the key contributions of this paper and their relations, covering i) general convergence properties, ii) the DFE and single-virus endemic equilibria, and iii) coexistence equilibria.

our system, we provide two algebraic characterizations for strong connectivity (Proposition 1 and Corollary 1). We identify a sufficient condition for two single-virus endemic equilibrium (corresponding to each of the two viruses) to be simultaneously unstable (Proposition 3). A class of system parameters that admits at least one coexistence equilibrium, when two single-virus endemic equilibria (one each corresponding to each of the two viruses) are unstable, has been identified in (Cui et al., 2025, Theorem 9). We provide a novel proof for a special case of (Cui et al., 2025, Theorem 9); see the proof of Proposition 4. Figure 1 provides a summary of the key contributions and how they relate to each other.

Some of the results in this paper have appeared in the Proceedings of the 2024 American Control Conference (Gracy et al., 2024). Note that Proposition 1, Corollary 1, Corollary 2, Lemma 5, Theorem 1 (and hence Lemma 7), Theorem 2 (and hence Lemma 2), Theorems 3 and 4, and Propositions 2, 5 and 6 were not included in (Gracy et al., 2024).

Structure of the Paper

The paper is organized as follows. We present relevant notation, and recall certain graph-theoretic notions in the rest of this section. We introduce the competitive networked bivirus SIS model with HOI, and formally state the problems of interest in Section 2. Our main results are split across Section 3, 4 and 5. Simulations to illustrate our theoretical findings are in Section 6. Finally, we conclude our paper in Section 7.

1.1. Notations and some preliminaries on linear algebra

We denote the set of real numbers by $\mathbb{R}$, and the set of nonnegative real numbers by $\mathbb{R}_+$. For any positive integer n , we use $[n]$ to denote the set $\{1, 2, \dots, n\}$. The i^{th} entry of a vector x is denoted by x_i . The element in the i^{th} row and j^{th} column of a matrix M is denoted by M_{ij} . We use $\mathbf{0}$ and $\mathbf{1}$ to denote the vectors whose entries all equal 0 and 1, respectively, and use I to denote the identity matrix, while the sizes of the vectors and matrices are to be understood from the context. For a vector x we denote

Table 1: Summary of key variables and their description

Var.	Description
$x_i^k(t)$	Fraction of node i infected with virus $k \in \{1, 2\}$ at time t
a_{ij}^k	Corresponds to a pairwise edge for virus k from node j to node i .
$b_{ij\ell}^k$	Corresponds to an HOI edge for virus k , from nodes j and node ℓ , to node i .
β_1^k	Infection rate of virus k due to pairwise edges
β_2^k	Infection rate of virus k due to HOI edges
δ_i^k	Recovery rate of node i from virus k .
$A^k = [a_{ij}^k]$	Matrix comprised of the entries a_{ij}^k
$B_i^k = [b_{ij\ell}^k]$	Matrix comprised of the entries $b_{ij\ell}^k$ model.
$D^k = \text{diag}(\delta_i^k)$	Diagonal matrix of the recovery rates for virus k

the diagonal square matrix with x along the diagonal by $\text{diag}(x)$. For any two real vectors $a, b \in \mathbb{R}^n$ we write $a \geq b$ if $a_i \geq b_i$ for all $i \in [n]$, $a > b$ if $a \geq b$ and $a \neq b$, and $a \gg b$ if $a_i > b_i$ for all $i \in [n]$. Likewise, for any two real matrices $A, B \in \mathbb{R}^{n \times m}$, we write $A \geq B$ if $A_{ij} \geq B_{ij}$ for all $i \in [n]$, $j \in [m]$, and $A > B$ if $A \geq B$ and $A \neq B$. For a square matrix M , we use $\sigma(M)$ to denote the spectrum of M , $\rho(M)$ to denote the spectral radius of M , and $s(M)$ to denote the largest real part among the eigenvalues of M , i.e., $s(M) = \max\{\text{Re}(\lambda) : \lambda \in \sigma(M)\}$. We denote a subset by $P \subseteq Q$, a proper subset by $P \subset Q$, and a set difference by $P \setminus Q$.

A real square matrix A is called Metzler if all its off-diagonal entries are nonnegative. If $A (= [a_{ij}]_{n \times n})$ is a nonnegative matrix, then $\rho(A)$ decreases monotonically with a decrease in a_{ij} for any $i, j \in [n]$. The matrix A is reducible if, and only if, there is a permutation matrix P such that $P^\top A P$ is block upper triangular; otherwise, A is said to be irreducible. If a nonnegative A is irreducible, and $Ax = y$ for $x > \mathbf{0}$, then $y > \mathbf{0}$, and y cannot have a zero in every position where x has a zero. A summary of the key notations used in the present paper is given in Table 1.

1.2. Graph theoretical notions

The notion of hypergraphs will play a crucial role in our analysis. Here, we recall certain concepts related to hypergraphs that will be needed in the sequel.

Note that the hypergraphs of interest in this paper have two important characteristics: they are directed, and the *head vertex set* of any hyperedge is always a single vertex; the latter is a consequence of the restriction imposed in this paper that an individual in node i can simultaneously interact with individuals belonging to at most two different nodes, say node j and node ℓ . For such a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the vertex set, and $\mathcal{E}$ is the set of hyperedges, each hyperedge $e \in \mathcal{E}$ is defined by an ordered pair $e = (S, v)$ where S is a set of *tail vertices* and $v \in \mathcal{V}$ is a *head vertex*. In fact, each hyperedge consists of either a simple edge joining two vertices or a pair of edges leaving from two possibly different vertices but terminating in a single vertex. For the special case where all tail vertex sets have one element, the notions of directed hypergraph and directed graph coincide.

Following the nomenclature in (Ausiello and Laura, 2017), we say that a *hyperpath* from a set of vertices S_1 to a single vertex t is an alternating sequence of vertices and hyperedges $\mathcal{P} = \{S_1, e_1, v_2, e_2, v_3, \dots, e_k, t\}$ such that (identifying t with v_{k+1})

- i) For each $j = 1, 2, \dots, k$, there holds $e_j = (S_j, v_{j+1})$ for some S_j satisfying $S_j \subseteq S_1 \cup \{v_2, v_3, \dots, v_{j-1}\}$, and
- ii) For each $j = 2, \dots, k+1$, there holds $v_j \in \cup_{j=2}^\ell S_j$.

Thus, each hyperedge used in the hyperpath must be a hyperedge of $\mathcal{H}$, all the tail vertices of a given hyperedge must have been reached previously in the hyperpath, and the head vertex is in the tail vertex set of another hyperedge. A directed hypergraph is strongly connected when, between any two vertices, there is a hyperpath.

It is clear that this is an equivalence relation for the vertices, and paralleling the situation applicable for a directed graph, the vertices of a hypergraph can be partitioned into subsets, with each subset constituting vertices that are maximally strongly connected. Further, from such a partition, a condensation graph (Bullo, 2024, Section 3.3.1) can be constructed, with each vertex corresponding to one maximal strongly connected set of vertices of the original hypergraph, and with an edge directed from one vertex of the condensation graph, call it a , to another, call it b , when the corresponding vertex sets S_a and S_b of the original hypergraph have the property that there is a hyperedge from S_a to any vertex in S_b . The condensation graph is an acyclic directed graph.

As such, and assuming the original hypergraph is not strongly connected, the condensation graph has more than one node, and the condensation graph has at least one leaf node. Consequently, in the hypergraph, there is one maximally strongly connected set of vertices from which no other vertex, outside that set, is reachable. If the vertex set is denoted by S_0 , there is no vertex v outside S_0 such that for H_0 a subset of S_0 , (H_0, v) is an edge of the hypergraph.

2. Problem Formulation

2.1. Model

Consider a network of n nodes. A node represents a well-mixed¹ population of individuals; the size of the population is fixed. We suppose that there are two viruses, say virus 1 and virus 2, spreading over such a network. Throughout this paper, we will assume that the two aforementioned viruses are competing with each other, which implies that no individual can be simultaneously infected with virus 1 and with virus 2. Through pairwise or higher-order interactions (HOI) (which we describe in more detail below), a healthy individual in node i gets infected with virus 1 (resp. virus 2) due to contact with either other

individuals who are infected with virus 1 (resp. virus 2). When, at a given time instant, a single interaction is involved (i.e., between two individuals in node i or between an individual in node i and an individual in node j), we say that the infection is due to *pairwise interactions*. An individual in node i might also be infected with virus 1 (resp. virus 2) due to *simultaneous* interactions with infected individuals in nodes j and ℓ , where, with j being possibly equal to ℓ either a) $j = i$, and/or $\ell = i$, or b) j, ℓ are neighbors of i . Such interactions are referred to as *higher-order interactions* (HOI).

We assume that the pairwise infection (resp. HOI) rate with respect to virus k is the same for all nodes, i.e., β_1^k (resp. β_2^k) for all $i \in [n]$ and $k \in [2]$. An individual infected with virus k , depending on its healing rate δ_i^k , recovers from said infection and becomes susceptible to again being infected by virus 1 or by virus 2. For a given node, all individuals in it have the same healing rate with respect to virus k ; individuals in different nodes possibly have different healing rates. Node i is said to be healthy if all individuals in node i are healthy; otherwise, we say it is infected. Note that, within the same node, it is possible for there to exist simultaneously i) a fraction that is infected with virus 1, and ii) a different fraction that is infected with virus 2.

The spread of the two competing viruses, with the possibility of HOI factored in, can be modeled using a 2-layer hypergraph, say $\mathcal{G}$. The vertices of $\mathcal{G}$ represent the population nodes. The k^{th} layer denotes the contact hypergraph for the spread of virus k , where $k = 1, 2$. More specifically, for each layer of $\mathcal{G}$, there exists a hyperedge corresponding to i) each pairwise interaction between individuals in nodes j and i , and ii) each HOI jointly between individuals in node j and individuals in node ℓ , and individuals in node i . Let E^k denote the hyperedge set corresponding to the k^{th} layer of $\mathcal{G}$. Let A^k (where $a_{ij}^k \geq 0$) denote the matrix, the entries of which capture the strength of pairwise interactions with respect to virus k . For each node i , let $B_i^k = [b_{ij\ell}^k]$ denote the matrix whose entries capture the strength of HOI with respect to virus k . The rows and columns of B_i^k are indexed by j and ℓ , which are the tail vertices of the hyperedge linking those vertices to vertex i . Clearly, the matrix must be symmetric. It is, however, easier when $j \neq \ell$, to take each of $(B_i^k)_{j\ell}$ and $(B_i^k)_{\ell j}$ to be one-half the total strength of the interaction attributable to the hyperedge. To summarize, for the k^{th} layer of the hypergraph $\mathcal{G}$, there exists a directed hyperedge from node j to node i or from nodes j, ℓ to node i if, and only if, $a_{ij}^k > 0$ or $b_{ij\ell}^k > 0$. Note that β_i^k can be considered the base infection rate for virus k for pairwise ($i = 1$) and HOI ($i = 2$) interactions, while a_{ij}^k and $b_{ij\ell}^k$ are parameters that scale this base rate. We choose this notation for convenience and to be consistent with past literature concerning HOI models (Cisneros-Velarde and Bullo, 2021; Cui et al., 2025, 2024), but point out that our formulation allows complete heterogeneity in infection rates for each node, in line with classical SIS bivirus liter-

¹Well-mixed means that the probability of any two individuals in a node interacting with each other is the same.

ature (Liu et al., 2019).

We use $x_i^k(t)$ to denote the fraction of individuals infected with virus k in node i at time instant t . The evolution of this fraction can, therefore, be represented by the following scalar differential equation (Cui et al., 2025, Section 5). Specifically, for $i = 1, 2, \dots, n$, we have

$$\begin{aligned}\dot{x}_i^1 &= -\delta_i^1 x_i^1 + \beta_1^1(1 - x_i^1 - x_i^2) \sum_{j=1}^n a_{ij}^1 x_j^1 + \\ &\quad \beta_2^1(1 - x_i^1 - x_i^2) \sum_{j,\ell=1}^n b_{ij\ell}^1 x_j^1 x_\ell^1 \\ \dot{x}_i^2 &= -\delta_i^2 x_i^2 + \beta_1^2(1 - x_i^1 - x_i^2) \sum_{j=1}^n a_{ij}^2 x_j^2 + \\ &\quad \beta_2^2(1 - x_i^1 - x_i^2) \sum_{j,\ell=1}^n b_{ij\ell}^2 x_j^2 x_\ell^2\end{aligned}\quad (1)$$

Define $D^1 = \text{diag}(\delta_i^1)$, where $i = 1, 2, \dots, n$, and define D^2 analogously. Define $X^1 = \text{diag}(x_i^1)$, where $i = 1, 2, \dots, n$, and define X^2 analogously. The matrices A^k and B_i^k for each $i \in [n]$ and $k \in [2]$ are as defined previously. Let $x^k = [x_1^k \ x_2^k \ \dots \ x_n^k]^\top$ for $k = 1, 2$. Therefore, in vector form, equation (1) can be written as:

$$\begin{aligned}\dot{x}^1 &= -D^1 x^1 + \beta_1^1(I - X^1 - X^2)A^1 x^1 + \\ &\quad \beta_2^1(I - X^1 - X^2)((x^1)^\top B_1^1 x^1, (x^1)^\top B_2^1 x^1, \dots, (x^1)^\top B_n^1 x^1)^\top \\ \dot{x}^2 &= -D^2 x^2 + \beta_1^2(I - X^1 - X^2)A^2 x^2 + \\ &\quad \beta_2^2(I - X^1 - X^2)((x^2)^\top B_1^2 x^2, (x^2)^\top B_2^2 x^2, \dots, (x^2)^\top B_n^2 x^2)^\top\end{aligned}\quad (2)$$

Note that $(x^\top B_1 x, x^\top B_2 x, \dots, x^\top B_n x)^\top = \begin{bmatrix} x^\top B_1 x \\ x^\top B_2 x \\ \vdots \\ x^\top B_n x \end{bmatrix}$.

Note that the model in (2) is indeed a compartmental model; it has three compartments S (susceptible), I^1 (infected with virus 1) and I^2 (infected with virus 2) that any individual can belong to, but the individuals are themselves identified with a specific population (node) in a network setting. The presence of the HOI terms allows for multiple interaction mechanisms for a susceptible individual to transition into either of the infected compartments.

We will sometimes refer to the hypergraph $\mathcal{G}$ described above (constructed from the matrices A^k , B_i^k) as the hypergraph *associated* with the system (2).

We will need the Jacobian of the right side of (2), and we note for this purpose, using the symmetric nature of B_i , that

$$\frac{\partial}{\partial x_j}(x^\top B_i x) = 2x^\top B_i e_j$$

and so

$$\nabla \begin{bmatrix} x^\top B_1 x \\ x^\top B_2 x \\ \vdots \\ x^\top B_n x \end{bmatrix} = 2 \begin{bmatrix} x^\top B_1 \\ x^\top B_2 \\ \vdots \\ x^\top B_n \end{bmatrix}. \quad (3)$$

We note that system (2) is a special case of (Cui et al., 2025, system 20) in the following sense: System (2) only accounts for the case where virus 1 (resp. virus 2) spread only due to contact with the infected individuals. In contrast, the model in (Cui et al., 2025) (see (Cui et al., 2025, system 20)) allows for the possibility of the viruses to spread through additional mediums such as a water distribution network, a public transit network, etc.

Remark 1 Note that setting $\beta_2^k = 0$ for $k = 1, 2$ results in system (2) coinciding with the classic networked bivirus SIS model, see (Ye and Anderson, 2022) and references therein. Setting, for all $t \in \mathbb{R}_{\geq 0}$, $x^1(t) = \mathbf{0}$ (resp. $x^2(t) = \mathbf{0}$) results in system (2) coinciding with the model used for studying the spread of a single virus over hypergraphs in (Cisneros-Velarde and Bullo, 2021).

The Jacobian of system (2) evaluated at an arbitrary point, (x^1, x^2) , in the state space is as given in (4).

$$J(x^1, x^2) = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}, \quad (4)$$

where

$$J_{11} = -D^1 + \beta_1^1(I - X^1 - X^2)A^1 - \text{diag}(\beta_1^1 A^1 x^1) + \beta_2^1(I - X^1 - X^2)O_1(x^1) - \beta_2^1 \text{diag}((x^1)^\top B_i^1 x^1)_{i=1,2,\dots,n} \quad (5)$$

$$J_{12} = -\text{diag}(\beta_1^1 A^1 x^1) - \beta_2^1 \text{diag}((x^1)^\top B_i^1 x^1)_{i=1,2,\dots,n} \quad (6)$$

$$J_{21} = -\text{diag}(\beta_1^2 A^2 x^2) - \beta_2^2 \text{diag}((x^2)^\top B_i^2 x^2)_{i=1,2,\dots,n} \quad (7)$$

$$J_{22} = -D^2 + \beta_1^2(I - X^1 - X^2)A^2 - \text{diag}(\beta_1^2 A^2 x^2) + \beta_2^2(I - X^1 - X^2)O_2(x^2) - \beta_2^2 \text{diag}((x^2)^\top B_i^2 x^2)_{i=1,2,\dots,n} \quad (8)$$

The terms $O_1(x^1)$ and $O_2(x^2)$ are as given in (9), being obtained from (3):

$$O_1(x^1) = 2 \begin{bmatrix} (x^1)^\top B_1^1 \\ (x^1)^\top B_2^1 \\ \vdots \\ (x^1)^\top B_n^1 \end{bmatrix} \quad O_2(x^2) = 2 \begin{bmatrix} (x^2)^\top B_1^2 \\ (x^2)^\top B_2^2 \\ \vdots \\ (x^2)^\top B_n^2 \end{bmatrix} \quad (9)$$

We need the following assumption so as to ensure that the model is well-defined.

Assumption 1 The matrix D^k , for $k = 1, 2$, is a positive diagonal matrix. The matrices A^k , for $k = 1, 2$, and B_i^k for $i \in [n]$ and $k \in [2]$ are nonnegative.

We define the set $\mathcal{D}$ as follows:

$$\mathcal{D} := \{(x^1, x^2) \mid x^k \geq \mathbf{0}, k = 1, 2, \sum_{k=1}^2 x^k \leq \mathbf{1}\}. \quad (10)$$

It is known that the origin is always an equilibrium for system (2), and, under Assumption 1, the set $\mathcal{D}$ is positively invariant; see (Cui et al., 2025, Lemma 3). The fact that $\mathcal{D}$ is positively invariant guarantees that the state values $x_i^k, k \in [2], i \in [n]$, stay in the $[0, 1]$ interval for all time instants. Given that the states represent fractions of a population node that is infected, values outside the $[0, 1]$ interval do not correspond to physical reality.

The model in system (2) has three kinds of equilibria:

- i) healthy state or disease-free equilibrium (DFE),
- ii) $(\mathbf{0}, \mathbf{0})$; single-virus endemic equilibria corresponding to virus k of the form $(\bar{x}^k, \mathbf{0})$, where $\mathbf{0} \ll \bar{x}^k \ll \mathbf{1}$ for $k = 1, 2$, and
- iii) coexisting equilibria, $(\bar{x}^1, \bar{x}^2)$, where, as we will show in Lemma 2, $\mathbf{0} \ll \bar{x}^1, \bar{x}^2 \ll \mathbf{1}$, and $\bar{x}^1 + \bar{x}^2 \ll \mathbf{1}$

We will use the terms “single-virus endemic equilibria” and “boundary equilibria” interchangeably. Note that it is not known if the single-virus endemic equilibria corresponding to virus k are necessarily unique; the papers (Cisneros-Velarde and Bullo, 2021; Cui et al., 2025) identify conditions for the existence of a single-virus endemic equilibria corresponding to virus k , but remain silent on the question of uniqueness.

2.2. Strong connectivity for hypergraphs

Recall that a directed hypergraph is strongly connected when, between any two vertices, there is a hyperpath. In this subsection, we will provide an algebraic characterization of the notion of strong connectedness in hypergraphs, and state and prove several properties of the matrices associated with a hypergraph that would be useful in proving some of the main results of this paper. For ease of exposition, we recall the single-virus networked SIS HOI model, which, as noted previously, is a special case of the model in (2).

$$\begin{aligned} \dot{x}(t) = & -Dx(t) + \beta_1(I - X)Ax(t) \\ & + \beta_2(I - X)(x(t)^\top B_1x(t), x(t)^\top B_2x(t), \dots, x(t)^\top B_nx(t))^\top \end{aligned} \quad (11)$$

Let G denote the hypergraph associated with system (11); note that the hypergraph association is the same as stated above for system (2), except here we consider just a single layer.

With the graph G defined, the following proposition gives an algebraic characterization of the notion of strong connectedness in hypergraphs.

Proposition 1 *Consider system (11). The following two conditions are equivalent:*

- i) *The hypergraph G is not strongly connected.*
- ii) *There exists a strict subset S_0 of the vertex set $\mathcal{V}$, assumed to have $|S_0| = n_0 < n$, such that if the vertices are ordered so that all vertices in S_0 appear before the remaining $n_1 = n - n_0$ vertices, then the matrices A and B_i have the following structural properties:*

$$A = \begin{bmatrix} A_{00} & A_{01} \\ 0 & A_{11} \end{bmatrix} \quad (12)$$

and for all $i \notin S_0$, i.e., for all $i \in \{n_0 + 1, \dots, n\}$,

$$B_i = \begin{bmatrix} 0 & (B_i)_{01} \\ (B_i)_{10} & (B_i)_{11} \end{bmatrix} \quad (13)$$

Proof: Suppose that the hypergraph G is not strongly connected. Construct S_0 as described in subsection 1.2. If the A_{10} (i.e., 2-1) block of A had a nonzero element, this would indicate the existence of a simple edge from one vertex in S_0 to a vertex in $\mathcal{V} \setminus S_0$, which would contradict the defining property of S_0 that there is no join of a single vertex in S_0 to a vertex outside S_0 . Similarly, if the $(B_i)_{00}$ or 1-1 block of B_i had a single nonzero entry for some $i \notin S_0$, this would indicate the existence of a vertex pair of S_0 with a hyperedge to a vertex i outside S_0 . Hence the

zero entries of A and B_i are a necessary consequence of the lack of strong connectedness of the hypergraph. The argument is very straightforward to reverse, to complete the demonstration of the equivalence of the two conditions. ■

Proposition 1 is a generalization to hypergraphs of the well-known connection between the notion of irreducibility of the adjacency matrix and strong connectivity of a directed graph. There is an additional property of nonnegative irreducible matrices that is useful to generalize to the hypergraph context. Recall first that a nonnegative matrix A is irreducible if and only if for any nonnegative and nonzero vector x , the vector Ax has at least one nonzero entry in the same position as where x has a zero entry (see Section 1.1). The following is a generalization of this idea.

Lemma 1 *Consider system (11). Let x be an arbitrary nonnegative but nonzero n -vector and suppose that $j_1, j_2, \dots, j_{n_1}$ denote the indices of the zero entries of x . Then under the condition that the entries $(Ax)_{j_k}$ of Ax and the scalars $x^\top B_{j_k}x$ for all $k = 1, 2, \dots, n_1$ are all zero, the hypergraph is not strongly connected. Conversely, if the hypergraph is not strongly connected, there exists such a nonnegative x such that the nominated quantities are zero.*

Proof: Suppose there exists a nonnegative but nonzero x for which the condition of the Lemma statement is fulfilled. Reorder the entries of x and correspondingly the rows and columns of A and the B_i , including the indexing of this matrix set, so that the nonzero entries of x occur before all the zero entries. If the last n_1 entries of Ax are all zero, it follows that A has the form of (12). Next, notice that $x^\top B_i x$ is nonzero precisely if the subblock matrix of B_i formed from its first n_0 rows and columns is nonzero. So if $x^\top B_j x$ is zero for all $j = n_0 + 1, n_0 + 2, \dots, n$, equation (13) holds for all such B_j , i.e., the lack of strong connectivity is established. Conversely, suppose there is a lack of strong connectivity. With the structure defined in Proposition 1, any x with the last n_1 entries zero will satisfy the conditions of the lemma. ■

The following corollary is almost immediate by negation of Lemma 1.

Corollary 1 *Consider system (11). The hypergraph G is strongly connected if, and only if, for any nonnegative, nonzero x for which entries $j_1, \dots, j_{n_1}$ constitute the set of zero entries, either the corresponding entries of Ax must contain one that is nonzero and/or one of the $x^\top B_{j_k}x$ must be nonzero*

Next, we turn our attention to showing that the irreducibility of a matrix including A and the B_i (defined as in system (11)) is a consequence of the strong connectivity of the hypergraph. Indeed, we have the following corollary:

Corollary 2 *Consider system (11) and suppose that the hypergraph G is strongly connected. Let each w_i , for $i =$*

$1, 2, \dots, n$, be a strictly positive vector. Then for any positive

β_1, β_2 , the matrix $\beta_1 A + \beta_2 \begin{bmatrix} w_1^\top B_1 \\ w_2^\top B_2 \\ \vdots \\ w_n^\top B_n \end{bmatrix}$ is irreducible

Proof: We shall prove the negation of the claim, by showing that if irreducibility fails, then the hypergraph is not strongly connected. Accordingly, suppose there exists a nonzero $x > \mathbf{0}$ such that if the zero entries of x are in positions $j_1, j_2, \dots, j_{n_1}$, then the corresponding entries $(Ax)_{j_k}$ of Ax are zero as are all $w_{j_k}^\top B_{j_k} x$. Without loss of generality, reorder the vertex numbering so that vertices $j_1, j_2, \dots, j_{n_1}$ become vertices $n_0 + 1, n_0 + 2, \dots, n$. This implies that the matrix A satisfies (12), while, for $i = n_0 + 1, n_0 + 2, \dots, n$, the matrix B_i satisfies (13). Therefore, by Proposition 1, it follows that the hypergraph is not strongly connected.

Notice that if either or both of the two matrices appearing in the statement of Corollary 2 are multiplied on the left by a diagonal matrix with positive diagonal entries, the conclusion of Corollary 2 remains true.

In light of the above, we adopt the following assumption in the present paper.

Assumption 2 *The hypergraph $\mathcal{G}^k$ for each $k \in [2]$ is strongly connected.*

In an epidemiological context, Assumption 2 implies that, for each of the two viruses, there exists an infection pathway from any node to any other node. An infection pathway is a hyperpath, indicating that the virus can be carried by infectious individuals from the starting node, through the sequence of hyperedges, to reach susceptible individuals at the end node of the hyperpath. In SIS network models on directed graphs, strong connectivity is often invoked (Liu et al., 2019; Doshi et al., 2022; Sahneh and Scoglio, 2014); it is a reasonable assumption from a graph-theoretic perspective. Our assumption extends this to the context of hypergraphs, but this (i.e., Assumption 2) does not imply that the pairwise interaction matrices A^k , $k = 1, 2$, are irreducible - in fact, Assumption 2 is weaker. A major contribution of this paper is to show that certain properties associated with system (2) which have so far been established under the stronger assumption that pairwise interaction matrices A^k , $k = 1, 2$, are irreducible (Cisneros-Velarde and Bullo, 2021; Cui et al., 2025) can be established using only Assumption 2.

2.3. Problem Statements

With respect to system (2), we aim to answer the following questions in this paper:

- i) What is the typical behavior that the trajectories exhibit as time goes to infinity?

- ii) Can a parameter regime be identified such that multiple (three, to be precise) equilibria are simultaneously (locally) stable?
- iii) Supposing that $x^2(t) = \mathbf{0}$ for all t (resp. $x^1(t) = \mathbf{0}$ for all t), provide a sufficient condition for local exponential convergence to a single-virus endemic equilibrium of virus 1 (resp. virus 2).
- iv) Supposing that the reproduction number of each virus is greater than one, identify sufficient conditions for the existence of a coexistence equilibrium. Furthermore, establish stability properties of such an equilibrium based on knowledge of the stability properties of the boundary equilibria.
- v) Identify a sufficient condition which guarantees that no coexistence equilibrium can exist.

2.4. Preliminary Lemmas

In this subsection, we secure certain preliminary results on the nature of equilibria of system (1), and recall certain known results on irreducible matrices. These results will play a crucial role in establishing the main findings of this paper.

The following lemma restricts the kinds of equilibria that system (2) may possess. The proof is partly inspired from (Ye et al., 2022, Lemma 3.1).

Lemma 2 *Consider system (2) under Assumptions 1 and 2. If $\bar{x} = (\bar{x}^1, \bar{x}^2) \in \mathcal{D}$ is an equilibrium of (2), then, for each $k \in [2]$, either $\bar{x}^k = \mathbf{0}$, or $\mathbf{0} \ll \bar{x}^k \ll \mathbf{1}$. Moreover, $\sum_{k=1}^2 \bar{x}^k \ll \mathbf{1}$.*

Proof: It is clear that $(\mathbf{0}, \mathbf{0})$ is an equilibrium of (2). Therefore, in the rest of the proof, we will show that any non-zero equilibrium $\bar{x} = (\bar{x}^1, \bar{x}^2)$ of (2) must satisfy, for each $k \in [2]$, $\mathbf{0} \ll \bar{x}^k \ll \mathbf{1}$ and $\sum_{k=1}^2 \bar{x}^k \ll \mathbf{1}$. We start off by showing that $\bar{x}^1 + \bar{x}^2 \ll \mathbf{1}$. For any $i \in [n]$, observe that the following is satisfied:

$$\begin{aligned} \dot{\bar{x}}_i^1 + \dot{\bar{x}}_i^2 = & -\delta_i^1 \bar{x}_i^1 - \delta_i^2 \bar{x}_i^2 + \beta_1(1 - \bar{x}_i^1 - \bar{x}_i^2) \sum_{j=1}^n a_{ij}^1 \bar{x}_j^1 \\ & + \beta_2(1 - \bar{x}_i^1 - \bar{x}_i^2) \sum_{j,\ell=1}^n b_{ij\ell}^1 \bar{x}_j^1 \bar{x}_\ell^1 \\ & + \beta_1^2(1 - \bar{x}_i^1 - \bar{x}_i^2) \sum_{j=1}^n a_{ij}^2 \bar{x}_j^2 \\ & + \beta_2^2(1 - \bar{x}_i^1 - \bar{x}_i^2) \sum_{j,\ell=1}^n b_{ij\ell}^2 \bar{x}_j^2 \bar{x}_\ell^2 \end{aligned} \quad (14)$$

Suppose that, for some $i \in [n]$, $\bar{x}_i^1 + \bar{x}_i^2 = 1$. Therefore, since, by Assumption 1, $\delta_i^k > 0$ for $k = 1, 2$, and since $\bar{x}_i^k \in \mathcal{D}$, from (14), it is clear that $\dot{\bar{x}}_i^1 + \dot{\bar{x}}_i^2 < 0$. However, since by assumption $\bar{x} = (\bar{x}^1, \bar{x}^2)$ is an equilibrium, it must be that $\dot{\bar{x}}_i^1 + \dot{\bar{x}}_i^2 = 0$, which is a contradiction. Therefore, for all $i \in [n]$, $\bar{x}_i^1 + \bar{x}_i^2 < 1$, which implies that $\sum_{k=1}^2 \bar{x}^k \ll \mathbf{1}$; thus guaranteeing that $\bar{x}^k \ll \mathbf{1}$ for $k = 1, 2$.

We are left to show that $\bar{x}^k \gg \mathbf{0}$ for $k = 1, 2$. To this end, suppose that $\bar{x}^1 > \mathbf{0}$ is an equilibrium point for which there exists at least one (but possibly more) $i \in [n]$ such

that $\bar{x}_i^1 = 0$. Note that the equilibrium version of the first line of equation (2) yields the following:

$$\begin{aligned} \dot{\bar{x}}^1 = & -D^1 \bar{x}^1 + \beta_1^1 (I - \bar{X}^1 - \bar{X}^2) A^1 \bar{x}^1 \\ & + \beta_2^1 (I - \bar{X}^1 - \bar{X}^2) ((\bar{x}^1)^\top B_1^1 \bar{x}^1, (\bar{x}^1)^\top B_2^1 \bar{x}^1, \dots, (\bar{x}^1)^\top B_n^1 \bar{x}^1)^\top \end{aligned} \quad (15)$$

By noting that $\bar{x}^1$ is an equilibrium point, and by a suitable rearrangement of terms, we obtain:

$$\bar{x}^1 = S \bar{x}^1, \quad (16)$$

where

$$\begin{aligned} S = & (D^1)^{-1} \beta_1^1 (I - \bar{X}^1 - \bar{X}^2) A^1 + \\ & (D^1)^{-1} \beta_2^1 (I - \bar{X}^1 - \bar{X}^2) ((\bar{x}^1)^\top B_1^1, \dots, (\bar{x}^1)^\top B_n^1)^\top. \end{aligned} \quad (17)$$

By Assumption 2, we know that the hypergraph is strongly connected. Therefore, since, by assumption, $\bar{x}^1 > \mathbf{0}$, from Corollary 1, it follows that there exists $i \in [n]$ such that if $\bar{x}_i^1 = 0$, then either a) $(A^1 \bar{x}^1)_i > 0$ and/or b) $(\bar{x}^1)^\top B_i^1 \bar{x}^1 > 0$. Therefore, since S is a nonnegative matrix (which follows from Assumption 1 and the fact that $I - \bar{X}^1 - \bar{X}^2$ is a positive diagonal matrix), there exists $i \in [n]$ such that $\bar{x}_i^1 = 0$ yet $(S \bar{x}^1)_i > 0$. Note that this contradicts (16). Hence, if $\bar{x}^1 > \mathbf{0}$ is an equilibrium point, then it must be that $\bar{x}^1 \gg \mathbf{0}$. By an analogous argument, it can be shown that if $\bar{x}^2 > \mathbf{0}$ is an equilibrium point, then it must be that $\bar{x}^2 \gg \mathbf{0}$, thus completing the proof. ■

The following lemmas will be needed for establishing (in)stability of different kinds of equilibria of system (2).

Lemma 3 (Varga, 1999, Lemma 2.3) *Suppose that M is an irreducible Metzler matrix. Then $r = s(M)$ is a simple eigenvalue of M , with an eigenvector $\zeta \gg \mathbf{0}$.*

Lemma 4 (Varga, 1999, Theorem 2.7) *Suppose that N is an irreducible nonnegative matrix. Then,*

- (i) $r = \rho(N)$ is a simple eigenvalue of N .
- (ii) There is an eigenvector $\zeta \gg \mathbf{0}$ corresponding to the eigenvalue r .
- (iii) $x > \mathbf{0}$ is an eigenvector only if $Nx = rx$ and $x \gg \mathbf{0}$.
- (iv) If A is a nonnegative matrix such that $A < N$, then $\rho(A) < \rho(N)$.

Lemma 5 *Suppose that Λ is a negative diagonal matrix and N is a nonnegative matrix. Let M be the Metzler matrix such that $M = \Lambda + N$. Then, $s(M) < 0$ if and only if $\rho(-\Lambda^{-1}N) < 1$, and $s(M) > 0$ if and only if, $\rho(-\Lambda^{-1}N) > 1$.*

Proof: Let Q be a matrix of all 1's, and set $N_\epsilon = N + \epsilon Q$. Suppose that $\rho(-\Lambda^{-1}N) < 1$. Then for small enough $\epsilon > 0$, $\rho(-\Lambda^{-1}N_\epsilon) < 1$, due to continuity of the spectral radius with respect to the entries of the matrix. Since N_ϵ is irreducible, then (Liu et al., 2019, Proposition 1) implies that $s(M_\epsilon) < 0$, where $M_\epsilon = \Lambda + N_\epsilon$. The spectral abscissa is a monotone function of ϵ and, since $M \leq M_\epsilon$, it must be that $s(M) \leq s(M_\epsilon)$ and hence $s(M) < 0$. For the

other direction, suppose that $s(M) < 0$. Since $M < M_\epsilon$, we have that $s(M) \leq s(M_\epsilon)$. By choosing ϵ small enough, we obtain $s(M_\epsilon) \leq 0$. We separately consider the case where $s(M_\epsilon) < 0$ and the case where $s(M_\epsilon) = 0$. Suppose that $s(M_\epsilon) < 0$, which, given that M_ϵ is irreducible Metzler, from (Liu et al., 2019, Proposition 1) implies that $\rho(-\Lambda^{-1}N_\epsilon) < 1$. This, from Lemma 4, further implies that $\rho(-\Lambda^{-1}N) < 1$. Suppose that $s(M_\epsilon) = 0$. Therefore, from (Liu et al., 2019, Proposition 1), it must be that $\rho(-\Lambda^{-1}N_\epsilon) = 1$, which, from Lemma 4, further implies that $\rho(-\Lambda^{-1}N) < 1$.

Suppose that $\rho(-\Lambda^{-1}N) > 1$. By noting that $-\Lambda^{-1}N_\epsilon > -\Lambda^{-1}N$, and by applying Lemma 4 statement iv), it is clear that $\rho(-\Lambda^{-1}N_\epsilon) > 1$. Since $-\Lambda^{-1}N_\epsilon$ is nonnegative irreducible, from (Liu et al., 2019, Proposition 1) it must be that $s(M_\epsilon) > 0$. Then for small enough ϵ , $s(M) > 0$. By reversing the arguments, it can be shown that $s(M) > 0$ implies $\rho(-\Lambda^{-1}N) > 1$. ■

Lemma 5 generalizes (Liu et al., 2019, Proposition 1), in that it does not insist that the matrix N be irreducible.

Lemma 6 (Rantzer, 2011, Proposition 2) *Let $A \in \mathbb{R}^{n \times n}$ be Metzler. Then, A is Hurwitz if, and only if, there exists an $x \in \mathbb{R}^n$ such that $x \gg \mathbf{0}$ and $Ax \ll \mathbf{0}$.*

In classical SIS bivirus systems, there are certain limitations on the trajectories; those starting on the boundary of $\mathcal{D}$ are always inward pointing, see (Ye et al., 2022, Lemma 3.2). Hence, it is natural to ask if there are similar restrictions on the trajectories of system (2). We will address this question in the rest of this subsection. Moreover, we will be doing so *without* insisting on the matrices A^k , for $k = 1, 2$, being irreducible - this is a sharp contrast from the approach adopted in (Ye et al., 2022, Lemma 3.2). We do, however, invoke an assumption that the hypergraphs $\mathcal{G}^k$ are strongly connected.

Lemma 7 *Consider system (2) under Assumptions 1 and 2. Suppose that the matrices B_i^k for $i \in [n]$ and $k \in [2]$ are nonnegative. Suppose that the initial conditions satisfy i) $x^k(0) > \mathbf{0}$ for $k \in [2]$, and ii) $(x^1(0), x^2(0)) \in \mathcal{D}$. Then for all finite $t > 0$, $\mathbf{0} \ll x^k(t) \ll \mathbf{1}$ for $k \in [2]$, and $x^1(t) + x^2(t) \ll \mathbf{1}$.*

Proof: Define $z := \mathbf{1} - x^1 - x^2$ (and $Z = \text{diag}(I - X^1 - X^2)$), and $\hat{B}^i := \text{diag}(\beta_i^1 A^i x^i)$. Therefore, (2) can be rewritten as:

$$\begin{aligned} \dot{x}^i(t) = & -D^i x^i(t) + \hat{B}^i z(t) \\ & + \beta_2^i Z((x^1)^\top B_1^i x^1, (x^1)^\top B_2^i x^1, \dots, (x^1)^\top B_n^i x^1)^\top, i = 1, 2 \\ \dot{z}(t) = & D^1 x^1(t) + D^2 x^2(t) - [\hat{B}^1 + \hat{B}^2] z(t) \\ & - \beta_2^1 Z((x^1)^\top B_1^1 x^1, (x^1)^\top B_2^1 x^1, \dots, (x^1)^\top B_n^1 x^1)^\top \\ & - \beta_2^2 Z((x^2)^\top B_1^2 x^2, (x^2)^\top B_2^2 x^2, \dots, (x^2)^\top B_n^2 x^2)^\top \end{aligned} \quad (18)$$

Suppose that for some $\tau \in \mathbb{R}_{\geq 0}$, and for some $i \in [n]$, $z_i(\tau) = 0$. Since $z_i(\tau) = 1 - x_i^1(\tau) - x_i^2(\tau)$, it is immediate that either $x_i^1(\tau) \neq 0$ and/or $x_i^2(\tau) \neq 0$. Therefore, a) since D^1 and D^2 are positive diagonal matrices, and b) since, by assumption, $z_i(\tau) = 0$, from the second line in

equation (18), we have that $\dot{z}_i(\tau) > 0$. This implies that, for $t > \tau$ and $t - \tau$ sufficiently small, $z_i(t) > 0$. Note that in the above calculation, the choice of node i is arbitrary; hence, the argument above applies for all $i \in [n]$. Hence, we have that $z_i(t) > 0$ for all t and $i \in [n]$. As a result, we obtain $z(t) \gg \mathbf{0}$, which implies that $x^1(t) + x^2(t) \ll \mathbf{1}$. Hence, we have $x^k(t) \ll \mathbf{1}$ for $k \in [2]$.

We now turn to establishing that $\mathbf{0} \ll x^k(t)$ for all finite $t > 0$. Suppose that for some $t \in \mathbb{R}_{\geq 0}$, ℓ entries in the vector $x^1(t)$ equal zero. We label these entries as $i_1, i_2, \dots, i_\ell$. By Assumption 2, we know that the hypergraph is strongly connected. Therefore from Corollary 1, since $x^1(t) \geq \mathbf{0}$ for all $t \in \mathbb{R}_{\geq 0}$ (see (Cui et al., 2025, Lemma 5.1)), it follows that there exists $i \in [n]$ such that $[x^1(t)]_i = 0$ yet either $[A^1 x^1(t)]_i > 0$ and/or $(x^1(t))^\top B_i x^1(t) > 0$. We assume that this happens for the i_1^{th} entry in x^1 , i.e., $[x^1(t)]_{i_1} = 0$ yet $[A^1 x^1(t)]_{i_1} > 0$ and/or $((x^1(t))^\top B_{i_1}^1 x^1(t) > 0$. The evolution of the infection level for virus 1 in node i_1 is as follows:

$$\begin{aligned} \dot{x}_{i_1}^1(t) &= [-D^1 x^1(t)]_{i_1} + [\beta_1^1(I - X^1 - X^2)A^1 x^1(t)]_{i_1} \\ &\quad + \beta_2^1(1 - x^1(t)_{i_1} - x^2(t)_{i_1})x^1(t)^\top B_{i_1}^1 x^1(t) \\ &> 0, \end{aligned} \quad (19)$$

where the inequality in (19) follows by noting that $x_{i_1}(t) = 0$ by assumption, and either $[\beta_1^1 A^1 x^1(t)]_{i_1} > 0$ and/or $x^1(t)^\top B_{i_1}^1 x^1(t) > 0$ as discussed above. This means that there must exist some time instant t' , with $t' - t$ not too large, such that $x^1(t')$ has fewer than ℓ zero entries. Repeating the argument for all the other zero entries in the vector $x^1(t')$ (indeed, this can be done since the choice of node i_1 was arbitrary), we have that $x^1(t) \gg \mathbf{0}$. Analogously, we can prove that $x^2(t) \gg \mathbf{0}$, thus concluding the proof.

3. Monotone dynamical systems and competitive bivirus networked SIS models with HOI

In this section, we establish that the bivirus SIS HOI system is strongly monotone, and provide a general conclusion on the limiting dynamical behavior of the same.

Monotone dynamical systems (MDS) are a class of nonlinear systems that are highly appealing in the context of mathematical epidemiology; for an overview of MDS, see (Smith, 1988; Hirsch, 1988). This is mainly because of the fact that MDS, assuming generic parameter values,² have a finite number of equilibria, and converge to a (stable) equilibrium point for all initial conditions except those lying on a measure-zero set. The property of MDS is as follows. Suppose that $(x_A^1(0), x_A^2(0))$ and $(x_B^1(0), x_B^2(0))$ are two initial conditions in $\text{int}(\mathcal{D})$ satisfying i) $x_A^1(0) > x_B^1(0)$ and ii) $x_A^2(0) < x_B^2(0)$. Then, if for all $t \in \mathbb{R}_{\geq 0}$ (resp.

$t \in \mathbb{R}_{>0}$), i) $x_A^1(t) \geq x_B^1(t)$ (resp. $x_A^1(t) \gg x_B^1(t)$) and ii) $x_A^2(t) \leq x_B^2(t)$ (resp. $x_A^2(t) \ll x_B^2(t)$), then system (2) is monotone (resp. strongly monotone) (Hirsch, 1988). In simpler terms, MDS are endowed with a trajectory ordering property.

3.1. The bivirus HOI system is strongly monotone

The following is the first main result of this paper.

Theorem 1 *Consider system (2) under Assumptions 1 and 2. Then, system (2) is strongly monotone.*

Proof: First, recall that a system is monotone (resp. strongly monotone) if the associated Jacobian (or a suitably transformed version of it) is Metzler (resp. irreducible Metzler); see (Hirsch, 1985, p. 424). Consider the permutation matrix $P := \text{diag}(I_n, -I_n)$, and observe that $PJ(x^1, x^2)P = \begin{bmatrix} J_{11} & -J_{12} \\ -J_{21} & J_{22} \end{bmatrix}$. By Assumption 1 and Lemma 7, matrices $-J_{12}$ and $-J_{21}$, and the off-diagonal terms in J_{11} and J_{22} are nonnegative; thus, implying that $PJ(x^1, x^2)P$ is Metzler. We are left to show that $PJ(x^1, x^2)P$ is irreducible in the interior of the region of interest.

As a connectivity argument reveals, it is intuitively reasonable, and was argued in (Ye et al., 2022), that the required irreducibility property will hold provided that J_{11} and J_{22} are separately irreducible, and J_{12} and J_{21} are both nonzero. Note that the Jacobian is as given in (4). By disregarding certain diagonal summands in the expression for J_{11} , it is evident that the irreducibility of J_{11} is equivalent to the irreducibility of $\hat{J}_{11}$, with

$$\hat{J}_{11} = \beta_1^1(I - X^1 - X^2)A^1 + 2\beta_2^1(I - X^1 - X^2) \begin{bmatrix} (x^1)^\top B_1^1 \\ (x^1)^\top B_2^1 \\ \vdots \\ (x^1)^\top B_n^1 \end{bmatrix}. \quad (20)$$

In the interior of $\mathcal{D}$, the matrix $I - X^1 - X^2$ is nonsingular with positive diagonal elements. Hence the irreducibility follows using Corollary 2. The irreducibility of J_{22} follows in the same way.

To see that J_{12} (and by the same argument J_{21}) is nonzero, suppose that to the contrary, $J_{12} = \mathbf{0}$. This would imply that for some $x^1 \gg \mathbf{0}$, there holds $A^1 x^1 = 0$ and $(x^1)^\top B_i^1 x^1 = 0$ for all i . Together with Lemma 1, this implies that all entries of A^1 and all entries of all B_i^1 are zero, which contradicts Assumption 2. ■

It is known that under Assumption 1 and the assumption that the pairwise interaction matrices A^k are irreducible (see Assumption 3 in Section 4 for a formal statement), system (2) is strongly monotone; see (Cui et al., 2025, Theorem 10). Indeed, if there are no HOI (i.e., $\beta_2^k = 0$ for $k = 1, 2$), the assumption that A^k is irreducible plays a central role in establishing that classical bi-virus systems are strongly monotone; see (Ye et al., 2022, Theorem 3.6). Theorem 1 shows that for bivirus systems with HOI the irreducibility of A^k is *not* required for establishing that the system is strongly monotone.

²The term ‘‘generic’’ means the following: for all but a set of parameter values that has measure zero. An algebraic or semi-algebraic set defines this set of exceptional values.

3.2. Finiteness of the number of equilibria for generic parameter choices

In this section, we establish that, for generic parameter choices, system (2) has a finite number of equilibria, and furthermore that the associated Jacobian is nonsingular. Note that generic finiteness of the number of equilibria has already been established, by means of an algebraic geometry approach, in (Cui et al., 2025, Theorem 10), but (Cui et al., 2025, Theorem 10) does not establish that, for generic parameter choices, the associated Jacobian is nonsingular. The gap is filled here by use of a different and efficient approach for proving finiteness of equilibria; the tool is the Parametric Transversality Theorem from differential topology, see (Lee, 2013, see p. 145) and (Guillemin and Pollack, 2010, see p.68). We use arguments very much like those in (Anderson and Ye, 2023) (where we studied the classic bivirus SIS setting), but our proof is more involved vis-a-vis the proof in (Anderson and Ye, 2023) due to a) the introduction of the terms associated with the HOIs for each of the two viruses, and b) the fact that we rely only on Assumption 2. Essentially because the healthy equilibrium and the single-virus boundary equilibria can be conveniently studied using single-virus techniques, it is easily established that there are no continua of equilibria confined to any boundary, i.e., any continuum of equilibria necessarily includes a continuum of coexistence equilibria. We focus, therefore, on showing that for generic parameter values, such equilibria cannot exist. The main result is as follows:

Theorem 2 *Consider system (2) under Assumptions 1 and 2. With any fixed matrices A^k and nonnegative B_i^k , and the exclusion of a set of values for the entries of D^1, D^2 of measure zero, the number of coexistence equilibrium points is finite, with their associated vector field zero being nondegenerate, i.e., with the associated Jacobian nonsingular. Similarly, with any fixed D^1, D^2 , and the exclusion of a set of values for the entries of A^k, B^k of measure zero, the same properties of equilibrium points hold.*

Proof: See Appendix. ■

3.3. Generic convergence

Theorems 1 and 2 together guarantee that all the assumptions of Hirsch's generic convergence theorem (Hirsch, 1985) are fulfilled. This allows us to draw conclusions regarding the limiting behavior of system (2); we have the following result.

Theorem 3 *Consider system (2) under Assumptions 1 and 2. For all initial conditions $(x^1(0), x^2(0)) \in \mathcal{D}$ except possibly for a set of measure zero, the system (2) will converge to an equilibrium. If the system does not converge to an equilibrium, then it is on a nonattractive limit cycle.*

In words, Theorem 3 establishes that the typical behavior of system (2) is convergence to *some* equilibrium; this could be healthy, or (one of the possibly many) single-virus boundary equilibria, or a coexistence equilibrium. Thus, Theorem 3 answers question i) raised in Section 2.3. Given that Theorems 1-3 do not require the pairwise interaction digraph to be strongly connected, Theorem 3 (resp. Theorem 2) is more general than (Gracy et al., 2024, Theorem 2), (Cui et al., 2025, Theorem 10) (resp. (Gracy et al., 2024, Theorem 1)), since (Gracy et al., 2024, Theorem 2), (Cui et al., 2025, Theorem 10) (resp. (Gracy et al., 2024, Theorem 1)) require the pairwise interaction matrices A^k to be irreducible.

4. Existence and local stability of boundary equilibria

In this section, we identify a parameter regime that permits three equilibria of system (2) to be simultaneously locally exponentially stable. Thereafter, for a different parameter regime, we identify a condition for the existence and instability of a boundary equilibrium. Finally, for the case when there is only one virus circulating in the network, we identify a condition for the existence and local exponential stability of an endemic equilibrium.

Proposition 2 *Consider system (2) under Assumption 1, and $B_i^k \geq 0$ for all $i \in [n]$ and $k \in [2]$. Define, for $k = 1, 2$, $\mathbf{1}_{B^k} \in \{0, 1\}^n$ by $(\mathbf{1}_{B^k})_i = 1$ if $B_i^k \neq \mathbf{0}$; otherwise $(\mathbf{1}_{B^k})_i = 0$. Suppose that the following conditions are fulfilled for $k = 1, 2$:*

- a) $\rho(\beta_1^k (D^k)^{-1} A^k) < 1$, and
- b) $\min_{i \in [n] \text{ s.t. } B_i^k \neq \mathbf{0}} \left(\frac{\beta_1^k}{\delta_i^k} (A^k \mathbf{1}_{B^k})_i + \frac{\beta_2^k}{2\delta_i^k} \mathbf{1}_{B^k}^\top B_i^k \mathbf{1}_{B^k} \right) > 2$.

Then, the following statements are true:

- i) *The DFE is locally exponentially stable.*
- ii) *Under Assumption 2, there exist equilibria $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$ where for $k = 1, 2$, there holds $\bar{x}^k \gg \mathbf{0}$, with also $\bar{x}_i^k > \frac{1}{2}$ for $k = 1, 2$, for any i such that $B_i^k \neq \mathbf{0}$.*
- iii) *Under Assumption 2, any such equilibrium point $(\bar{x}^1, \mathbf{0})$ is locally exponentially stable.*
- iv) *Under Assumption 2, any such equilibrium point $(\mathbf{0}, \bar{x}^2)$ is locally exponentially stable.*

Proof of statement i): Note that the Jacobian evaluated at the DFE is as follows:

$$J(\mathbf{0}, \mathbf{0}) = \begin{bmatrix} -D^1 + \beta_1^1 A^1 & \mathbf{0} \\ \mathbf{0} & -D^2 + \beta_1^2 A^2 \end{bmatrix}. \quad (21)$$

By assumption, $\beta_1^k \rho((D^k)^{-1} A^k) < 1$, for $k = 1, 2$. Therefore, from Lemma 5, it must be that $s(-D^k + \beta_1^k A^k) < 0$ for $k = 1, 2$, which, since $J(\mathbf{0}, \mathbf{0})$ is a block diagonal matrix, and since the matrices $-D^1 + \beta_1^1 A^1$ and $-D^2 + \beta_1^2 A^2$ are the only blocks along the main diagonal, implies that $s(J(\mathbf{0}, \mathbf{0})) < 0$. Local exponential stability of the DFE,

then, follows from (Khalil, 2002, Theorem 4.15 and Corollary 4.3).

Proof of statement ii): Observe that the point $(\bar{x}^1, \mathbf{0})$, with $\bar{x}^1 \gg \mathbf{0}$, is an equilibrium of (2) (in fact, a single-virus endemic equilibrium of virus 1) if and only if $\bar{x}^1$ is an endemic equilibrium of (11). The same is true when we consider $(\mathbf{0}, \bar{x}^2)$. Thus, we focus on (11), and drop the superscript k from $\bar{x}$, A , B_i , D , and $\mathbf{1}_B$ for clarity. Our approach is the same as that in the proof of (Cisneros-Velarde and Bullo, 2021, Theorem 5.1, statement iv)); the crucial difference is that we establish the claim using Assumption 2, whereas (Cisneros-Velarde and Bullo, 2021, Theorem 5.1, statement iv)) requires a more restrictive assumption, namely that the matrix A be irreducible. First, we introduce the following functions: a) $h_+(z) = \frac{z}{1+z}$ for any $z \in \mathbb{R}_{\geq 0}$; and b) $H_+(y) = (h_+(y_1), h_+(y_2), \dots, h_+(y_n))^\top$ for $y \geq \mathbf{0}$. Define the map $H(x) := H_+(\bar{A}x + (x^\top \bar{B}_1 x, x^\top \bar{B}_2 x, \dots, x^\top \bar{B}_n x))$, where $\bar{A} := \beta_1 D^{-1} A$, and, for each $i \in [n]$, $\bar{B}_i := \frac{\beta_2}{\delta_i} B_i$. Clearly, a fixed point of map $H(\cdot)$ is an equilibrium of (11). Consider the vector $\mathbf{1}_B^\top$, as given in the statement of the proposition but particularized for the single-virus case. Define $Y := \{y \in [0, 1]^n \mid \frac{1}{2} \mathbf{1}_B \leq y \leq \mathbf{1}\}$. It turns out that $H(y) > \frac{1}{2} \mathbf{1}_B$. To see this, consider the following: For any $y \in Y$,

$$\begin{aligned} H(y) &= H_+(\bar{A}y + (y^\top \bar{B}_1 y, y^\top \bar{B}_2 y, \dots, y^\top \bar{B}_n y)^\top) \\ &\geq H_+(\bar{A} \frac{1}{2} \mathbf{1}_B + \frac{1}{4} ((\mathbf{1}_B)^\top \bar{B}_1 \mathbf{1}_B, \dots, (\mathbf{1}_B)^\top \bar{B}_n \mathbf{1}_B)^\top), \end{aligned} \quad (22)$$

where inequality (22) comes from the fact that a) $y \geq \frac{1}{2} \mathbf{1}_B$, and b) the function $h_+(\cdot)$ is monotonic in its argument. Observe that

$$\begin{aligned} \frac{1}{2} (\bar{A} \mathbf{1}_B)_i + \frac{1}{4} (\mathbf{1}_B)^\top \bar{B}_i \mathbf{1}_B &= \frac{1}{2} \frac{\beta_1}{\delta_i} \sum_{j=1}^n a_{ij} (\mathbf{1}_B)_j + \frac{1}{4} \frac{\beta_2}{\delta_i} (\mathbf{1}_B)^\top B_i \mathbf{1}_B \\ &= \frac{1}{4} \alpha_i, \end{aligned} \quad (23)$$

where $\alpha_i := 2 \frac{\beta_1}{\delta_i} \sum_{j=1}^n a_{ij} (\mathbf{1}_B)_j + \frac{\beta_2}{\delta_i} (\mathbf{1}_B)^\top B_i \mathbf{1}_B$. Notice that (23) holds for every $i \in [n]$, and that the left hand side of (23) is bounded from below by zero. By statement b) of the proposition hypothesis, it is clear that, for any i such that $B_i \neq 0$, $\alpha_i > 4$. This implies that $h_+(\frac{1}{4} \alpha_i) > \frac{1}{2}$. Consequently, we have that $H(y) \geq H_+(\frac{1}{4} \text{diag}(\alpha_i)_{i=1,2,\dots,n} \mathbf{1}_B) > \frac{1}{2} \mathbf{1}_B$.

Thanks to Assumption 1, we can be assured that $(\bar{A}y)_i + y^\top \bar{B}_i y \geq 0$, which implies that $h_+((\bar{A}y)_i + y^\top \bar{B}_i y) \leq 1$. This, in turn, implies that $H(y) \leq \mathbf{1}$. This, coupled with the fact that $H(y) > \frac{1}{2} \mathbf{1}_B$ as established above, means that $H : Y \rightarrow Y$. Therefore, using Brouwer's fixed-point theorem (Shapiro and Shapiro, 2016, Theorem 4.5), we conclude that there exists a fixed point $\bar{y} \in Y$ such that $H(\bar{y}) = \bar{y}$.

Next, we show that $\bar{y} \gg \mathbf{0}$. To this end, note that any fixed point of H , say $\bar{y}$, must satisfy the following

$$\frac{(\bar{A}\bar{y})_i + \bar{y}^\top \bar{B}_i \bar{y}}{1 + (\bar{A}\bar{y})_i + \bar{y}^\top \bar{B}_i \bar{y}} = \bar{y}_i, \quad i = 1, 2, \dots, n \quad (24)$$

Suppose, to obtain a contradiction, that $\bar{y}_{j_1}, \bar{y}_{j_2}, \dots, \bar{y}_{j_k}$ are the zero entries of $\bar{y}$. Then the left side of (24) is zero for $i = j_1, j_2, \dots, j_k$ and there must then hold

$$\bar{y}^\top \bar{B}_{j_1} \bar{y} = \bar{y}^\top \bar{B}_{j_2} \bar{y} = \dots = \bar{y}^\top \bar{B}_{j_k} \bar{y} = 0$$

and also

$$(\bar{A}\bar{y})_{j_1} = (\bar{A}\bar{y})_{j_2} = \dots = (\bar{A}\bar{y})_{j_k} = 0$$

By Corollary 1, it then follows that the hypergraph is not strongly connected, which contradicts the proposition hypothesis (namely, that Assumption 2 holds). This implies that $\bar{y} \gg \mathbf{0}$.

We will now show that $\bar{y}_i > \frac{1}{2}$ for i such that $B_i \neq \mathbf{0}$. To this end, recall that by statement b) of the proposition hypothesis, we have the following:

$$\min_{i \in [n] \text{ s.t. } B_i \neq \mathbf{0}} \left(\frac{\beta_1}{\delta_i} (A \mathbf{1}_B)_i + \frac{\beta_2}{2\delta_i} \mathbf{1}_B^\top B_i \mathbf{1}_B \right) > 2 \quad (25)$$

Now the mapping $y \rightarrow H(y)$ maps any $y \in Y$ with $\frac{1}{2} \mathbf{1}_B \leq y \leq \mathbf{1}_n$ into Y , and the equilibrium point $\bar{y}$ is contained in Y . Invoking the lower limit, we observe that for such a point,

$$(\bar{A}\bar{y})_i + \bar{y}^\top \bar{B}_i \bar{y} \geq (\bar{A} \frac{1}{2} \mathbf{1}_B)_i + \frac{1}{4} \mathbf{1}_B^\top \bar{B}_i \mathbf{1}_B \quad (26)$$

But in view of (25), the inequality in (26) means that $(\bar{A}\bar{y})_i + \bar{y}^\top \bar{B}_i \bar{y} > 1$; this, in combination with (24), ensures that $\bar{y}_i > \frac{1}{2}$.

Proof of statement iii): Our approach is similar to that in the proof of (Cisneros-Velarde and Bullo, 2021, Theorem 5.1, statement v)). Consider the equilibrium point $(\bar{x}^1, \mathbf{0})$, and observe that the Jacobian evaluated at this equilibrium is as follows:

$$J(\bar{x}^1, \mathbf{0}) = \begin{bmatrix} \bar{J}_{11} & \bar{J}_{12} \\ \mathbf{0} & \bar{J}_{22} \end{bmatrix}, \quad (27)$$

where

$$\begin{aligned} \bar{J}_{11} &= -D^1 + \beta_1^1 (I - \bar{X}^1) A^1 - \text{diag}(\beta_1^1 A^1 \bar{x}^1) + \\ &\quad \beta_2^1 (I - \bar{X}^1) O_1(\bar{x}^1) - \beta_2^1 \text{diag}((\bar{x}^1)^\top B_i^1 \bar{x}^1)_{i=1,2,\dots,n} \\ \bar{J}_{12} &= -\text{diag}(\beta_1^1 A^1 \bar{x}^1) - \beta_2^1 \text{diag}((\bar{x}^1)^\top B_i^1 \bar{x}^1)_{i=1,\dots,n} \\ \bar{J}_{22} &= -D^2 + \beta_1^2 (I - \bar{X}^1) A^2. \end{aligned}$$

The term $O_1(\bar{x}^1)$ is as defined in (9). We will establish the exponential stability of the 11 and 22 blocks (i.e., $\bar{J}_{11}$ and $\bar{J}_{22}$) separately. Observe that

$$\begin{aligned} \bar{J}_{11} &= -D^1 + \beta_1^1 (I - \bar{X}^1) A^1 - \text{diag}(\beta_1^1 A^1 \bar{x}^1) \\ &\quad + \beta_2^1 (I - \bar{X}^1) O_1(\bar{x}^1) - \beta_2^1 \begin{bmatrix} (\bar{x}^1)^\top B_1^1 \bar{x}^1 & & \\ & \ddots & \\ & & (\bar{x}^1)^\top B_n^1 \bar{x}^1 \end{bmatrix}. \end{aligned}$$

Define the summands

$$Q_1 := -D^1 + \beta_1^1(I - \bar{X}^1)A^1 + \beta_2^1(I - \bar{X}^1) \begin{bmatrix} (\bar{x}^1)^\top B_1^1 \\ \vdots \\ (\bar{x}^1)^\top B_n^1 \end{bmatrix}, \text{ and}$$

$$Q_2 := \beta_2^1(I - \bar{X}^1) \begin{bmatrix} (\bar{x}^1)^\top (B_1^1)^\top \\ \vdots \\ (\bar{x}^1)^\top (B_n^1)^\top \end{bmatrix} - \text{diag}(\beta_1^1 A^1 \bar{x}^1) - \beta_2^1 \begin{bmatrix} (\bar{x}^1)^\top B_1^1 \bar{x}^1 & & \\ & \ddots & \\ & & (\bar{x}^1)^\top B_n^1 \bar{x}^1 \end{bmatrix}.$$

It is immediate that $\bar{J}_{11} = Q_1 + Q_2$, which implies that $\bar{J}_{11}\bar{x}^1 = Q_1\bar{x}^1 + Q_2\bar{x}^1$. Since $\bar{x}^1$ is a single-virus endemic equilibrium corresponding to virus 1, by taking recourse to the equilibrium version of the first line of equation (2), it is clear that $Q_1\bar{x}^1 = \mathbf{0}$. Hence, $\bar{J}_{11}\bar{x}^1 = Q_2\bar{x}^1$.

Note that

$$Q_2\bar{x}^1 = \beta_2^1(I - \bar{X}^1) \begin{bmatrix} (\bar{x}^1)^\top (B_1^1)^\top \bar{x}^1 \\ \vdots \\ (\bar{x}^1)^\top (B_n^1)^\top \bar{x}^1 \end{bmatrix} - \text{diag}(\beta_1^1 A^1 \bar{x}^1) \bar{x}^1 - \beta_2^1 \begin{bmatrix} (\bar{x}^1)^\top B_1^1 \bar{x}^1 & & \\ & \ddots & \\ & & (\bar{x}^1)^\top B_n^1 \bar{x}^1 \end{bmatrix} \bar{x}^1. \quad (28)$$

Denote by $(Q_2\bar{x}^1)_i$ the i^{th} entry of the vector $Q_2\bar{x}^1$. Therefore, in view of (28), we have the following:

$$(Q_2\bar{x}^1)_i = -\beta_1^1 \left(\sum_{j=1}^n a_{ij}^1 \bar{x}_j^1 \right) \bar{x}_i^1 + \beta_2^1 (1 - 2\bar{x}_i^1) ((\bar{x}^1)^\top B_i^1 \bar{x}^1) \quad (29)$$

We consider its sign, under two circumstances.

Case 1: Suppose that $B_i^1 = \mathbf{0}$. Note that by Assumption 2 the hypergraph $\mathcal{G}$ is strongly connected, which, from Corollary 2 coupled with Lemma 7, implies that the matrix

$\beta_1^1 A^1 + \begin{bmatrix} (\bar{x}^1)^\top B_1^1 \\ (\bar{x}^1)^\top B_2^1 \\ \vdots \\ (\bar{x}^1)^\top B_n^1 \end{bmatrix}$ is irreducible. Consequently, the hypothesis that $B_i^1 = \mathbf{0}$ implies that the i^{th} row of matrix A^1 has a positive sum. This, since $\bar{x}_i^1 > 0$, implies that $\beta_1^1 \left(\sum_{j=1}^n a_{ij}^1 \bar{x}_j^1 \right) \bar{x}_i^1 > 0$. Thus, $(Q_2\bar{x}^1)_i < 0$.

Case 2: Suppose secondly that $B_i^1 \neq \mathbf{0}$. Therefore, from statement ii), it follows that $\bar{x}_i^1 > \frac{1}{2}$. Consequently, $1 - 2\bar{x}_i^1 < 0$; thus implying that $(Q_2\bar{x}^1)_i < 0$.

Note that, for the analysis above, the choice of index i was arbitrary; therefore, again, we have $(Q_2\bar{x}^1)_i < 0$ for all $i \in [n]$. Hence, since $\bar{J}_{11}\bar{x}^1 = Q_2\bar{x}^1$, it follows that $(\bar{J}_{11}\bar{x}^1)_i < 0$ for all $i \in [n]$. Note that Assumptions 1 and 2, together with Corollary 2 and Lemma 7 guarantee that the matrix $\bar{J}_{11}$ is an irreducible Metzler matrix; the reasons are as detailed in the proof of Theorem 1. Therefore, from Lemma 6, it must be that the matrix $\bar{J}_{11}$ is Hurwitz.

Turning to the 22 block of the Jacobian (i.e., matrix $\bar{J}_{22}$),

consider the matrices $\beta_1^2(D^2)^{-1}A^2$ and $\beta_1^2(D^2)^{-1}(I - \bar{X}^1)A^2$. From Assumption 1, it is clear that $\beta_1^2(D^2)^{-1}A^2$ is a nonnegative matrix. Since, from statement ii), $\bar{x}^1$ satisfies $\mathbf{0} \ll \bar{x}^1 \ll \mathbf{1}$, it is also clear that $\beta_1^2(D^2)^{-1}(I - \bar{X}^1)A^2$ is a nonnegative matrix. Furthermore, we immediately obtain $\beta_1^2(D^2)^{-1}(I - \bar{X}^1)A^2 < \beta_1^2(D^2)^{-1}A^2$. Therefore, since the spectral radius of a nonnegative matrix decreases monotonically with a decrease in any entry of said matrix (see Section 1.1), it follows that $\rho(\beta_1^2(D^2)^{-1}(I - \bar{X}^1)A^2) \leq \rho(\beta_1^2(D^2)^{-1}A^2)$. By assumption, $\rho(\beta_1^2(D^2)^{-1}A^2) < 1$, which implies that $\rho(\beta_1^2(D^2)^{-1}(I - \bar{X}^1)A^2) < 1$, and consequently, from Lemma 5, we have that $s(-D^2 + (I - \bar{X}^1)A^2) < 0$. Therefore, since $J(\bar{x}^1, \mathbf{0})$ is block upper triangular, and since we have already established that $\bar{J}_{11}$ is Hurwitz, it follows that $s(J(\bar{x}^1, \mathbf{0})) < 0$. Local exponential stability of $(\bar{x}^1, \mathbf{0})$, then, follows from (Khalil, 2002, Theorem 4.15 and Corollary 4.3).

Proof of statement iv): The proof is same as that of statement iii), up to a suitable adjustment of notation. ■

Proposition 2 answers question ii) raised in Section 2.3. Observe that Proposition 2 guarantees existence of boundary equilibria for the case when $\rho(\beta_1^k(D^k)^{-1}A^k) < 1$. It is natural to ask if one is assured of existence even if the spectral radius inequality was flipped. To answer this, we require a stronger assumption, which is as follows:

Assumption 3 The matrix A^k , for $k = 1, 2$, is irreducible.

The epidemiological interpretation of Assumption 3 remains the same as that of Assumption 2 with the exception that here, one is assured of the existence of an infection pathway even when higher-order interactions are not factored in (i.e., when considering just the graph of pairwise interactions). Assumption 3 is a standard assumption in mathematical epidemiology; see for instance (Liu et al., 2019; Doshi et al., 2022; Sahneh and Scoglio, 2014). With Assumption 3 in place, we have the following proposition.

Proposition 3 Consider system (2) under Assumptions 1 and 3. Suppose that, for all $k \in [2]$, $\rho(\beta_1^k(D^k)^{-1}A^k) > 1$. Then system (2) has at least three equilibria, namely the DFE, which is unstable; a single-virus endemic equilibrium corresponding to virus 1 $(\bar{x}^1, \mathbf{0})$; and a single-virus endemic equilibrium corresponding to virus 2 $(\mathbf{0}, \bar{x}^2)$. Furthermore, if $s(-D^i + \beta_1^i(I - \bar{X}^k)A^i) > 0$ for each $i, k \in [2]$ such that $i \neq k$, then the equilibrium points $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$ are unstable.

Proof: Observe that the DFE is always an equilibrium of system (2). Suppose that for some $k \in [2]$, $\rho(\beta_1^k(D^k)^{-1}A^k) > 1$, then from (21) it is clear that $s(J(\mathbf{0}, \mathbf{0})) > 0$. This, in view of (Khalil, 2002, Theorem 4.7, statement ii)), guarantees that the DFE is unstable. Further, from (Cisneros-Velarde and Bullo, 2021, Theorem 5.1, statement vii)) we know that there exists an endemic equilibrium, $\bar{x}^k$, where $\mathbf{0} \ll \bar{x}^k \ll \mathbf{1}$. Since, by assumption, $\rho(\beta_1^k(D^k)^{-1}A^k) > 1$

for all $k \in [2]$, it is also immediate that there exist equilibria, $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$, where $\mathbf{0} \ll \bar{x}^k \ll \mathbf{1}$, for $k = 1, 2$. Observe that the Jacobian evaluated at the point $(\bar{x}^1, \mathbf{0})$ is as given in (27). Further, note that $J(\bar{x}^1, \mathbf{0})$ is block upper triangular. By assumption, $s(-D^1 + \beta_1^1(I - \bar{X}^1)A^1) > 0$, which implies that $s(J(\bar{x}^1, \mathbf{0})) > 0$. Consequently, instability of $(\bar{x}^1, \mathbf{0})$ follows from (Khalil, 2002, Theorem 4.7, statement ii)). The instability of $(\mathbf{0}, \bar{x}^2)$ can be shown analogously, thus completing the proof. ■

Note that Proposition 3 does not say anything about the situation when one of the spectral radii is greater than one, and the other is less than one. We discuss this here. Suppose that $s(-D^2 + \beta_1^2(I - \bar{X}^1)A^2) > 0$ and $s(-D^1 + \beta_1^1(I - \bar{X}^2)A^1) < 0$. Then, since by assumption $s(-D^2 + \beta_1^2(I - \bar{X}^1)A^2) > 0$, from Proposition 3 it is clear that $(\bar{x}^1, \mathbf{0})$ is unstable. By assumption, $s(-D^1 + \beta_1^1(I - \bar{X}^2)A^1) < 0$. However, whether or not $s(-D^1 + \beta_1^1(I - \bar{X}^2)A^1) < 0$ guarantees (local exponential) stability of $(\mathbf{0}, \bar{x}^2)$ is an open question. So no conclusions can be drawn regarding the stability of $(\mathbf{0}, \bar{x}^2)$.

For the single-virus case, a sufficient condition for global stability of the endemic equilibrium has been identified in (Cisneros-Velarde and Bullo, 2021, Theorem 5.1, statement viii)). Regardless, the following caveat with respect to said result should be noted: (Cisneros-Velarde and Bullo, 2021, Theorem 5.1, statement viii) is reliant on that fact that, for a given β_1 , there exists a small enough β_2 such that for the resulting single-virus system the endemic equilibrium is globally exponentially stable. This is clearly *not* the case if one were to seek a sufficient condition where both the parameters β_1 and β_2 are arbitrary. It turns out, however, that one can identify a sufficient condition for local exponential attractivity of the endemic equilibrium in the single-virus case, even when the parameters β_1 and β_2 are arbitrary. We have the following theorem.

Theorem 4 *Consider system (11) under Assumptions 1 and 3. Suppose that the parameters associated with system (11) are generic. If $\beta_1 \rho(D^{-1}A) > 1$, then there exists an equilibrium point $\bar{x}$ such that $\bar{x} \gg \mathbf{0}$. Furthermore, the point $\bar{x}$ is asymptotically stable.*

Proof: First, observe that (Cui et al., 2025, Theorem 6) implies that system (11) is an irreducible monotone system, and that, for generic parameter choices, for almost all initial conditions, the dynamics converge to a stable equilibrium point. Note that for system (11) the origin is an equilibrium, and there are no other equilibria on the boundary of the set $[0, 1]^n$; see (Cisneros-Velarde and Bullo, 2021, Lemma 5.1, statement iii)). By assumption, $\beta_1 \rho(D^{-1}A) > 1$, which, from Lemma 5 implies that $s(-D + \beta_1 A) > 0$. Consequently, from the analysis in the proof of Proposition 2, it is clear that the origin is unstable. Since $\beta_1 \rho(D^{-1}A) > 1$, it follows from (Cisneros-Velarde and Bullo, 2021, Theorem 5.1, statement vii)) that there exists an equilibrium $\bar{x}$ such that $\bar{x} \gg \mathbf{0}$. Thus, all the hypotheses of (Smith, 1988, Theorem 2.8) are satisfied, and

consequently we conclude that $s(J(\bar{x})) \leq 0$. ■

Theorem 2 implies that, for generic parameter matrices, the Jacobian of system (11), at any equilibrium point, is nonsingular. This means that $J(\bar{x})$ cannot have an eigenvalue with zero real part and zero imaginary part. It can be easily verified that $J(\bar{x})$ is an irreducible Metzler matrix. Therefore, from Lemma 3, it follows that the eigenvalue with the largest real part *does not* have any imaginary parts. Consequently, the possibility that, for generic parameter choices, $s(J(\bar{x})) = 0$ can be ruled out. It follows that (Smith, 1988, Theorem 2.8, statement iv)), coupled with the fact that the set $\mathcal{D}$ is positively invariant (see (Cui et al., 2025, Lemma 3)), guarantees that the equilibrium point $\bar{x}$ attracts at least all initial conditions x such that $\mathbf{0} \leq x \ll \bar{x}$. Therefore, it is clear that $\bar{x}$ is locally asymptotically stable.

5. (Non)existence of Coexistence equilibria

This section deals with the establishment of sufficient conditions for the existence (resp. nonexistence) of coexistence equilibria. The questions we are tackling in this section are not just mere academic exercises. In fact, as has correctly been pointed out in (Ye and Anderson, 2022), knowledge of the conditions that lead to the coexistence of viruses is extremely useful, since health practitioners can leverage said knowledge to determine whether there is a need of drugs, antibiotics, health infrastructure, and trained personnel to deal with just one or both viruses in the long term.

We consider two parameter regimes, viz., for $k = 1, 2$, i) $s(-D^k + \beta_i^k A^k) > 0$, and ii) $s(-D^k + \beta_i^k A^k) < 0$, and show that for each of these, there exists at least one coexistence equilibrium.

Proposition 4 *Consider system (2) under Assumptions 1 and 3. Let $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$ denote a single-virus endemic equilibrium corresponding to virus 1 and virus 2, respectively. Suppose that the following conditions are satisfied:*

- i) $s(-D^1 + \beta_1^1 A^1) > 0$;
- ii) $s(-D^2 + \beta_1^2 A^2) > 0$;
- iii) $s(-D^1 + \beta_1^1(I - \bar{X}^2)A^1) > 0$; and
- iv) $s(-D^2 + \beta_1^2(I - \bar{X}^1)A^2) > 0$.

Then, there exists at least one equilibrium of the form $(\hat{x}^1, \hat{x}^2)$ such that $\mathbf{0} \ll \hat{x}^1, \hat{x}^2 \ll \mathbf{1}$ and $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$.

Prior to proving the claim in Proposition 4, we need the following background material. In line with the terminology of (Hofbauer, 1990), given an equilibrium point of system (2), it is classified as saturated or unsaturated. An equilibrium is saturated (resp. strictly saturated) if the diagonal block of the Jacobian corresponding to the zero entries of said equilibrium has no eigenvalues with positive real part (resp. has every eigenvalue to be strictly less

than zero) and unsaturated otherwise (Hofbauer, 1990). A boundary equilibrium of (2) is saturated if and only if said boundary equilibrium is locally exponentially stable; this follows immediately by noting the structure of the Jacobian matrix, evaluated at a boundary equilibrium, see (27). The definition also implies that every fixed point in the interior of $\mathcal{D}$, irrespective of its stability properties, is saturated (Hofbauer, 1990); therefore, from Lemma 2, we have that every coexistence equilibrium of system (2) is saturated.

Proof: Assumptions i) and ii) of Proposition 4 guarantee existence of boundary equilibria, $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$; see Proposition 3. Observe that (Cui et al., 2025, Lemma 3) guarantees that, for each $k \in [2]$, $x^k(0) \geq \mathbf{0}$ implies that $x^k(t) \geq \mathbf{0}$ for all $t \in \mathbb{R}_{\geq 0}$, and that the set $\mathcal{D}$ (which is compact) is forward invariant. Therefore, from (Hofbauer, 1990, Theorem 2), it follows that system (2) has at least one saturated fixed point. There are two cases to consider.

Case 1: Suppose that the aforementioned saturated fixed point is in the interior of $\mathcal{D}$. Note that any fixed point in the interior of $\mathcal{D}$ is of the form $(\hat{x}^1, \hat{x}^2)$, where $\mathbf{0} \ll (\hat{x}^1, \hat{x}^2) \ll \mathbf{1}$, thus implying that $(\hat{x}^1, \hat{x}^2)$ is a coexistence equilibrium. From Lemma 2, it must necessarily satisfy $\mathbf{0} \ll (\hat{x}^1, \hat{x}^2) \ll \mathbf{1}$, and $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$.

Case 2: Suppose, but we will demonstrate a contradiction, that there are no fixed points in the interior of $\mathcal{D}$. This implies that there must be a saturated fixed point on the boundary of $\mathcal{D}$ (Hofbauer, 1990). Therefore, at least one of the single-virus boundary equilibria is saturated.

However, from Proposition 3, it is clear that assumptions iii) and iv) guarantee that the boundary equilibria are unstable; thus implying that they are unsaturated, and the contradiction is obtained. ■

Proposition 4 is implied by (Cui et al., 2025, Theorem 9), which, assuming $\beta_2^k = 0$ for $k = 1, 2$, is the same as (Janson et al., 2020, Theorem 5), (Doshi et al., 2022, Theorem 5.4). The proof of (Cui et al., 2025, Theorem 9) is lengthy, since it primarily relies on fixed point mapping, Perron-Frobenius theory, etc. Our proof is significantly shorter, since we exploit the advantages that the notion of MDS endows our system with.

Note further that Proposition 4 makes no comment on the number of coexistence equilibria. This gap may be partially addressed as follows. Thanks to Theorem 2, one could perhaps leverage (Hofbauer, 1990, Theorem 2) so as to obtain a lower bound on the number of coexistence equilibria; this is very much the situation for classic bivirus networked SIS models; see (Anderson and Ye, 2023, Corollary 3.10, statement 1). Thereafter, since system (2) is monotone, the properties of MDS can be exploited to possibly conclude that there must exist a locally exponentially stable coexistence equilibrium. A detailed treatment of this issue is beyond the scope of the present paper.

For the same parameter regime as in Proposition 4, we now present a different condition that ensures the existence of a coexistence equilibrium point $(\hat{x}^1, \hat{x}^2)$, but, differently from the situation in Proposition 4, we assume that both

boundary equilibria are stable.

Theorem 5 *Consider system (2) under Assumptions 1 and 3. Suppose that the parameters associated with system (2) are generic. Let $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$ denote a single-virus endemic equilibrium corresponding to virus 1 and virus 2, respectively. Suppose that the following conditions are satisfied:*

- i) $s(-D^1 + \beta_1^1 A^1) > 0$;
- ii) $s(-D^2 + \beta_1^2 A^2) > 0$;

Suppose that both $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$ are locally exponentially stable. Then there exists at least one equilibrium of the form $(\hat{x}^1, \hat{x}^2)$ such that $\mathbf{0} \ll \hat{x}^1, \hat{x}^2 \ll \mathbf{1}$ and $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$, such that $(\hat{x}^1, \hat{x}^2)$ is unstable.

Proof: By assumption, $s(-D^k + \beta_1^k A^k) > 0$ for $k = 1, 2$. Therefore, from Proposition 3, it follows that there exists a single-virus endemic equilibrium corresponding to virus 1, $\bar{x}^1 \gg \mathbf{0}$, and a single-virus endemic equilibrium corresponding to virus 2, $\bar{x}^2 \gg \mathbf{0}$. By assumption, both $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$ are locally exponentially stable.

The condition $s(-D^1 + \beta_1^1 A^1) > 0$ implies that the origin is unstable; this can be observed from the proof of statement i) in Proposition 2. We are left to show that the stable manifold of the origin does not lie in the interior of $\mathcal{D}$. We will do so by relying on the proof technique of (Anderson and Ye, 2023, Lemma 3.8). It suffices to show that for the (linear) system

$$\begin{bmatrix} \dot{x}^1 \\ \dot{x}^2 \end{bmatrix} = \begin{bmatrix} -D^1 + \beta_1^1 A^1 & \mathbf{0} \\ \mathbf{0} & -D^2 + \beta_1^2 A^2 \end{bmatrix} \begin{bmatrix} x^1 \\ x^2 \end{bmatrix} \quad (30)$$

no trajectory starting in the interior of $\mathcal{D}$ converges to the origin. First, consider $x^1(t)$. Let $w^\top$ be the positive left eigenvector associated with $s(-D^1 + \beta_1^1 A^1)$ such that all its entries sum to one. Define $z := w^\top x^1$, and observe that $\dot{z} = w^\top \dot{x}^1$, which, from (30), further implies that $\dot{z} = s(-D^1 + \beta_1^1 A^1)z$. Since, by assumption, $s(-D^1 + \beta_1^1 A^1) > 0$, it is clear that the projection onto w (which is a positive vector) of the points of (30) in the interior of $\mathcal{D}$ is away from $x^1 = \mathbf{0}$. An analogous argument can be made for $x^2(t)$, since, by assumption, $s(-D^2 + \beta_1^2 A^2) > 0$. Therefore, the stable manifold of the origin does not lie in the interior of $\mathcal{D}$. Consequently, since we know that system (2) is monotone and the monotone condition $\bar{x}^1 \gg \mathbf{0}, \bar{x}^2 \gg \mathbf{0}$ relates the two exponentially stable equilibrium points, from (Smith, 1988, Proposition 2.9) it follows that there exists an equilibrium point of the form $(\hat{x}^1, \hat{x}^2)$ such that $\mathbf{0} \ll \hat{x}^1, \hat{x}^2 \ll \mathbf{1}$ and $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$. Furthermore, the point $(\hat{x}^1, \hat{x}^2)$ satisfies $s(J(\hat{x}^1, \hat{x}^2)) \geq 0$. From Theorem 2, we know that the Jacobian, at any equilibrium point, is nonsingular. This means that $J(\hat{x}^1, \hat{x}^2)$ cannot have an eigenvalue with zero real part and zero imaginary part. Observe that $J(\hat{x}^1, \hat{x}^2)$ is an irreducible Metzler matrix. Therefore, from Lemma 3, it follows that the eigenvalue with the largest real part *does not* have any imaginary parts. Consequently, $s(J(\hat{x}^1, \hat{x}^2)) > 0$, which implies that the equilibrium point $(\hat{x}^1, \hat{x}^2)$ is unstable. ■

Proposition 4 and Theorem 5 partially answer question iv) raised in Section 2.3.

We now turn our attention to a different parameter regime, namely $s(-D^k + \beta_1^k A^k) < 0$ for $k \in [2]$, and provide a sufficient condition which guarantees the existence of an unstable coexistence equilibrium. We have the following result.

Proposition 5 *Consider system (2) with generic parameter matrices, under Assumptions 1 and 2. Suppose further that for each $k \in [2]$ there exists some $i \in [n]$ such that $B_i^k \neq \mathbf{0}$. Define, for $k = 1, 2$, $\mathbf{1}_{B^k} \in \{0, 1\}^n$ by $(\mathbf{1}_{B^k})_i = 1$ if $B_i^k \neq \mathbf{0}$; otherwise $(\mathbf{1}_{B^k})_i = 0$. Suppose that the following conditions are fulfilled for $k = 1, 2$:*

- a) $\rho(\beta_1^k (D^k)^{-1} A^k) < 1$, and
- b) $\min_{i \in [n] \text{ s.t. } B_i^k \neq \mathbf{0}} \left(\frac{\beta_1^k}{\delta_i^k} (A^k \mathbf{1}_{B^k})_i + \frac{\beta_2^k}{2\delta_i^k} \mathbf{1}_{B^k}^\top B_i^k \mathbf{1}_{B^k} \right) > 2$.

Then there exists at least one equilibrium of the form $(\hat{x}^1, \hat{x}^2)$ such that $\mathbf{0} \ll \hat{x}^1, \hat{x}^2 \ll \mathbf{1}$ and $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$ that is unstable.

Proof: Suppose that the conditions in Proposition 5 are fulfilled. Then, it follows that there exists boundary equilibria $(\hat{x}^1, \mathbf{0})$ and $(\mathbf{0}, \hat{x}^2)$, and that both are locally exponentially stable; see statements iii) and iv) in Proposition 2. Therefore, since we know that system (2) is monotone, from (Smith, 1988, Proposition 2.9) it follows that there exists (at least) one equilibrium point of the form $(\hat{x}^1, \hat{x}^2)$ such that $\mathbf{0} \ll \hat{x}^1, \hat{x}^2 \ll \mathbf{1}$ and $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$. The rest of the proof (i.e., showing that $(\hat{x}^1, \hat{x}^2)$ is unstable) is similar to the approach in the proof of Theorem 5. ■

Proposition 5 assures us of the existence of at least one coexistence equilibrium. Recall that system (2) is monotone. Therefore, since Theorem 2 says that, for generic parameter matrices, for each of the equilibrium points the associated Jacobian is nonsingular, the conditions in Proposition 5 can be shown to guarantee the existence of an odd number of coexistence equilibria, each of which must be unstable. The proof follows from a Brouwer degree argument; see proof of (Smith, 1986, Proposition 3.7). In fact, for classical bivirus SIS systems, for the same stability configuration as in Theorem 5 and Proposition 5, a lower bound on the number of coexistence equilibria has been obtained in (Anderson and Ye, 2023, Corollary 3.10, statement 2).

We now present a necessary condition for the existence of a coexistence equilibrium. We define for $k = 1, 2$ the following matrices: $R^k := [B_1^k \mathbf{1} \ B_2^k \mathbf{1} \ \dots \ B_n^k \mathbf{1}]^\top$. We have the following proposition.

Proposition 6 *Consider system (2) under Assumptions 1 and 2. Suppose that, for all $k \in [2]$, $\rho(\beta_1^k (D^k)^{-1} A^k) > 1$. Suppose that $D^k = I$ for $k = 1, 2$. If $\beta_1^1 A^1 + \beta_2^1 R^1 < \beta_1^2 A^2$, then there exists no coexistence equilibrium.*

Proof: Suppose that $\beta_1^1 A^1 + \beta_2^1 R^1 < \beta_1^2 A^2$. Assume by way of contradiction that there exists a coexisting equilibrium, $(\hat{x}^1, \hat{x}^2)$ such that $\mathbf{0} \ll (\hat{x}^1, \hat{x}^2) \ll \mathbf{1}$, and $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$.

Hence, since by assumption $D^k = I$ for $k = 1, 2$, by considering the equilibrium version of equation (2), we have the following:

$$-\hat{x}^1 + \beta_1^1 (I - \hat{X}^1 - \hat{X}^2) A^1 \hat{x}^1 + \beta_2^1 (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^1)^\top B_1^1 \hat{x}^1 \\ (\hat{x}^1)^\top B_2^1 \hat{x}^1 \\ \vdots \\ (\hat{x}^1)^\top B_n^1 \hat{x}^1 \end{bmatrix} = \mathbf{0} \quad (31)$$

$$-\hat{x}^2 + \beta_1^2 (I - \hat{X}^1 - \hat{X}^2) A^2 \hat{x}^2 + \beta_2^2 (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^2)^\top B_1^2 \hat{x}^2 \\ (\hat{x}^2)^\top B_2^2 \hat{x}^2 \\ \vdots \\ (\hat{x}^2)^\top B_n^2 \hat{x}^2 \end{bmatrix} = \mathbf{0} \quad (32)$$

From (31) and (32), we have the following:

$$(\beta_1^1 (I - \hat{X}^1 - \hat{X}^2) A^1 + \beta_2^1 (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^1)^\top B_1^1 \\ (\hat{x}^1)^\top B_2^1 \\ \vdots \\ (\hat{x}^1)^\top B_n^1 \end{bmatrix}) \hat{x}^1 = \hat{x}^1 \quad (33)$$

$$(\beta_1^2 (I - \hat{X}^1 - \hat{X}^2) A^2 + \beta_2^2 (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^2)^\top B_1^2 \\ (\hat{x}^2)^\top B_2^2 \\ \vdots \\ (\hat{x}^2)^\top B_n^2 \end{bmatrix}) \hat{x}^2 = \hat{x}^2 \quad (34)$$

By Assumption 1, we know that the matrices A^1 and A^2 are nonnegative and $\beta_1^1, \beta_2^1, \beta_1^2$ and β_2^2 are all strictly positive, and $(I - \hat{X}^1 - \hat{X}^2)$ is a positive diagonal matrix (because by Lemma 2, $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$), whereas, because of Assumption 2, from Corollary 2 we have that the matrix

$$\beta_1^k (I - \hat{X}^1 - \hat{X}^2) A^k + \beta_2^k (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^k)^\top B_1^k \\ (\hat{x}^k)^\top B_2^k \\ \vdots \\ (\hat{x}^k)^\top B_n^k \end{bmatrix} \text{ is ir-}$$

reducible. Therefore, it follows that the matrices $\beta_1^1 (I - \hat{X}^1 - \hat{X}^2) A^1 + \beta_2^1 (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^1)^\top B_1^1 \\ (\hat{x}^1)^\top B_2^1 \\ \vdots \\ (\hat{x}^1)^\top B_n^1 \end{bmatrix}$ and $\beta_1^2 (I - \hat{X}^1 - \hat{X}^2) A^2 + \beta_2^2 (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^2)^\top B_1^2 \\ (\hat{x}^2)^\top B_2^2 \\ \vdots \\ (\hat{x}^2)^\top B_n^2 \end{bmatrix}$ are nonnegative ir-

reducible. Since $\hat{x}^1, \hat{x}^2 \gg \mathbf{0}$, from Lemma 4, it follows that

$$\rho(\beta_1^1 (I - \hat{X}^1 - \hat{X}^2) A^1 + \beta_2^1 (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^1)^\top B_1^1 \\ (\hat{x}^1)^\top B_2^1 \\ \vdots \\ (\hat{x}^1)^\top B_n^1 \end{bmatrix}) = 1 \quad (35)$$

and

$$\rho(\beta_1^2 (I - \hat{X}^1 - \hat{X}^2) A^2 + \beta_2^2 (I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^2)^\top B_1^2 \\ (\hat{x}^2)^\top B_2^2 \\ \vdots \\ (\hat{x}^2)^\top B_n^2 \end{bmatrix}) = 1 \quad (36)$$

By assumption, $\beta_1^1 A^1 + \beta_2^1 R^1 < \beta_1^2 A^2$. Since $\hat{x}^1 \ll \mathbf{1}$, it must be that $\beta_1^1 A^1 + \beta_2^1 \begin{bmatrix} (\hat{x}^1)^\top B_1^1 \\ (\hat{x}^1)^\top B_2^1 \\ \vdots \\ (\hat{x}^1)^\top B_n^1 \end{bmatrix} < \beta_1^1 A^1 + \beta_2^1 R^1 < \beta_1^2 A^2$.

Since $(I - \hat{X}^1 - \hat{X}^2)$ is positive diagonal, we obtain the

following:

$$\beta_1^1(I - \hat{X}^1 - \hat{X}^2)A^1 + \beta_2^1(I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^1)^\top B_{B_1^1}^1 \\ (\hat{x}^1)^\top B_{B_2^1}^1 \\ \vdots \\ (\hat{x}^1)^\top B_{B_n^1}^1 \end{bmatrix} < \beta_1^1(I - \hat{X}^1 - \hat{X}^2)A^2 \quad (37)$$

Applying the condition in statement iv) of Lemma 4 to (37) yields:

$$\rho(\beta_1^1(I - \hat{X}^1 - \hat{X}^2)A^1 + \beta_2^1(I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^1)^\top B_{B_1^1}^1 \\ (\hat{x}^1)^\top B_{B_2^1}^1 \\ \vdots \\ (\hat{x}^1)^\top B_{B_n^1}^1 \end{bmatrix}) < \rho(\beta_1^1(I - \hat{X}^1 - \hat{X}^2)A^2) \quad (38)$$

From (35) and (38), $\rho(\beta_1^1(I - \hat{X}^1 - \hat{X}^2)A^2) > 1$. Therefore, since $\beta_2^2 > 0$, $(I - \hat{X}^1 - \hat{X}^2)$ is positive diagonal and the matrix $\begin{bmatrix} (\hat{x}^2)^\top B_{B_1^2}^2 \\ (\hat{x}^2)^\top B_{B_2^2}^2 \\ \vdots \\ (\hat{x}^2)^\top B_{B_n^2}^2 \end{bmatrix}$ is nonnegative, it is clear that

$$\rho(\beta_1^2(I - \hat{X}^1 - \hat{X}^2)A^2 + \beta_2^2(I - \hat{X}^1 - \hat{X}^2) \begin{bmatrix} (\hat{x}^2)^\top B_{B_1^2}^2 \\ (\hat{x}^2)^\top B_{B_2^2}^2 \\ \vdots \\ (\hat{x}^2)^\top B_{B_n^2}^2 \end{bmatrix}) > 1, \quad (39)$$

which contradicts (36). Therefore, there does not exist a coexistence equilibrium $(\hat{x}^1, \hat{x}^2)$, where $\mathbf{0} \ll (\hat{x}^1, \hat{x}^2) \ll \mathbf{1}$ and $\hat{x}^1 + \hat{x}^2 \ll \mathbf{1}$; thus completing the proof. ■

Proposition 6 answers question v) raised in Section 2.3, and highlights a situation where one virus dominates the other, referred to as *competitive exclusion*. It can also be shown that the condition in Proposition 6 implies that a) the DFE is unstable, and b) the boundary equilibrium, $(\bar{x}^1, \mathbf{0})$ is unstable.

Proposition 6 leads to an observation in the context of control of epidemics as discussed in the following remark.

Remark 2 *The presence of HOI makes it harder for one virus to achieve competitive exclusion. To see this, note that, for the case without HOI, a sufficient condition for one virus to dominate the other is $\beta_1^1 A^1 < \beta_1^2 A^2$; see (Ye et al., 2022, Corollary 3.11, statement 1). In contrast, for the case with HOI, Proposition 6 establishes that a sufficient condition for one virus to dominate the other is $\beta_1^1 A^1 + \beta_2^1 R^1 < \beta_1^2 A^2$. Since $\beta_1^1 A^1 + \beta_2^1 R^1 < \beta_1^2 A^2$ implies $\beta_1^1 A^1 < \beta_1^2 A^2$, but the converse is false, it follows that it is harder to achieve competitive exclusion in a bivirus network with HOI. Therefore, in the context of two strains of a virus, where one is benign and the other is malignant, and the control strategy is to have the benign strain dominate the malignant strain, the presence of HOI makes control of epidemics more challenging.*

6. Numerical Examples

In this section, we present two sets of numerical simulations in order to showcase different aspects of the theoretical contributions. In Section 6.1, we use $n = 4$ networks

with simple and easily interpretable structures, that satisfy Assumption 2. This allows one to clearly understand changes to the structure (including HOIs) and the infection rates can impact the system dynamics. We then consider in Section 6.2 a large-scale ($n = 107$) real-world network with a complex and nontrivial structure. This larger network satisfies Assumption 3 but not Assumption 2.

6.1. A simple 4-node network

We present a series of simulations of the bivirus system with HOIs to highlight interesting phenomena that can emerge when HOIs are incorporated. The network has $n = 4$ nodes, and we set $D^1 = D^2 = I$. The pairwise interactions are captured by the following matrices

$$A^1 = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad A^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The HOI are captured by the following nonzero B_i^k matrices (all other $B_i^k = 0$):

$$B_1^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B_4^1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$B_1^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad B_4^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

First, note that A^k is not irreducible for $k = 1, 2$, and so the graph associated with just the pairwise interactions is not strongly connected (Assumption 3 is not satisfied). Nonetheless, the hypergraph associated with A^k and $B_i^k, i = 1, 2, \dots, n$ is strongly connected for $k = 1, 2$ (Assumption 2 is satisfied).

In our simulations, we randomly sample $x_i^k(0)$ from a uniform distribution $(0, 1)$, and then normalize the vectors $x^1(0)$ and $x^2(0)$ to ensure that $(x^1(0), x^2(0)) \in \text{int}(\mathcal{D})$. The β_i^k are varied to yield different stability properties for the system in (2).

Example 1: We set $\beta_1^1 = \beta_2^1 = 0.2$ and $\beta_1^2 = \beta_2^2 = 5$. This ensures the inequalities of both conditions for Proposition 2 are satisfied. For initial conditions close to the DFE (Fig. 2a), the trajectories converge to the locally exponentially stable DFE, $(x^1 = \mathbf{0}, x^2 = \mathbf{0})$. When the initial conditions are further in the interior of $\mathcal{D}$, and depending on the particular initial condition, we observe convergence to one of two boundary equilibrium (both of which are locally exponentially stable), $(\bar{x}^1, \mathbf{0})$ or $(\mathbf{0}, \bar{x}^2)$ for some positive $\bar{x}^1 > 0.5 \times \mathbf{1}$ and $\bar{x}^2 > 0.5 \times \mathbf{1}$, see Figs. 2b and 2c, respectively. It is important to note that, without HOIs, it is impossible for a bivirus system to have the DFE, $(\bar{x}^1, \mathbf{0})$, and $(\mathbf{0}, \bar{x}^2)$ all locally exponentially stable (Liu et al., 2019)

Example 2: We set $\beta_1^1 = \beta_1^2 = 2$, and $\beta_2^1 = \beta_2^2 = 0.5$. The resulting bivirus system satisfies the first two inequality conditions in Proposition 4. The system also has two boundary equilibria $(\bar{x}^1, \mathbf{0})$ and $(\mathbf{0}, \bar{x}^2)$, which satisfy the

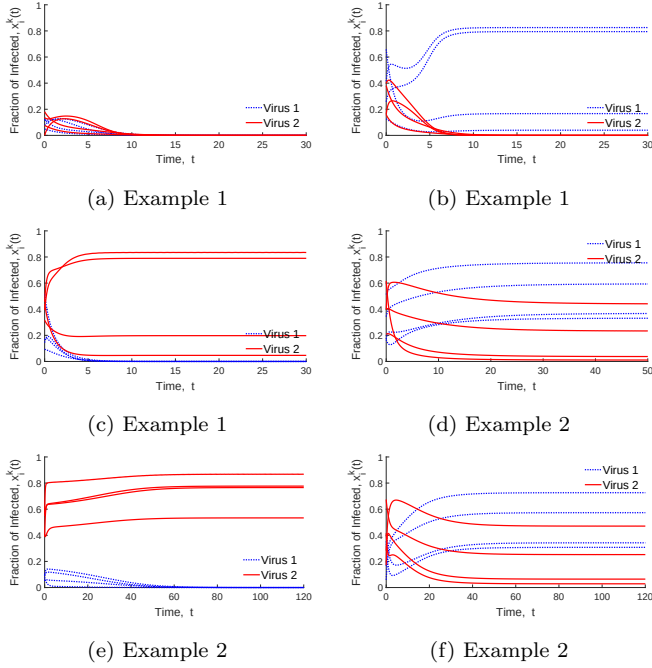


Figure 2: Trajectories of the system (2), for different simulation parameters and initial conditions. 2a, 2b and 2c correspond to different initial conditions for *Example 1*. 2d, 2e, and 2f correspond to *Example 2*, for different HOI interaction strengths for virus 2, and different initial conditions.

last two inequality conditions in Proposition 4. Proposition 3 indicates that the two boundary equilibria are unstable, and this can indeed be verified. As noted below Proposition 3, there is no theoretical proof that $(\bar{x}^1, \mathbf{0})$ (resp. $(\mathbf{0}, \bar{x}^2)$) is the unique boundary equilibrium for virus 1 (resp. virus 2) but simulations suggest the uniqueness property holds true. Figure 2d indicates that the system converges to a coexistence equilibrium, $(\hat{x}^1, \hat{x}^2)$ for some $\hat{x}^1 \gg \mathbf{0}$ and $\hat{x}^2 \gg \mathbf{0}$; the presence of such a coexistence equilibrium is predicted by Proposition 4 (though the Proposition is silent on the stability properties of any coexistence equilibrium). Additional simulations suggest that this coexistence equilibrium is the unique attractor of all initial conditions in the interior of $\mathcal{D}$.

Next, we increase the HOI interaction strength for virus 2, by changing $\beta_2^2 = 0.7$. Fig. 2e and 2f show convergence to different equilibria for different initial conditions (i.e., there are multiple locally exponentially stable equilibria). Importantly, of the two stable equilibria observed, one is a boundary equilibria $(\mathbf{0}, \bar{x}^2)$, where virus 2 is endemic and virus 1 extinct, while the other is a coexistence equilibrium. Note that the inequality $s(-D^2 + \beta_1^2(I - \bar{X}^1)A^2) > 0$ of Proposition 3 is violated for the boundary equilibrium $(\mathbf{0}, \bar{x}^2)$, but as noted below the proposition, this is not enough to theoretically establish the local exponential stability of the boundary equilibrium $(\mathbf{0}, \bar{x}^2)$; this contrasts with the bivirus system without HOIs (Ye et al., 2022).

Example 3: We set $\beta_1^1 = 2.5$, $\beta_2^1 = 2$, $\beta_1^2 = 2$ and $\beta_2^2 = 2.5$. The inequalities of Theorem 5 are satisfied.

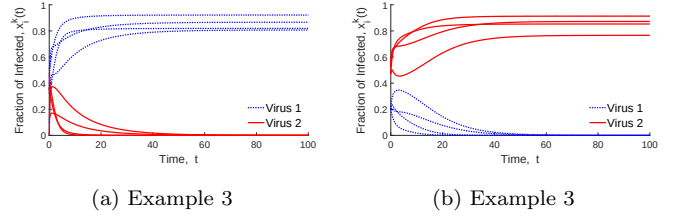


Figure 3: Trajectories of the system (2), for different simulation parameters and initial conditions. 3a and 3b correspond to different initial conditions for *Example 3*.

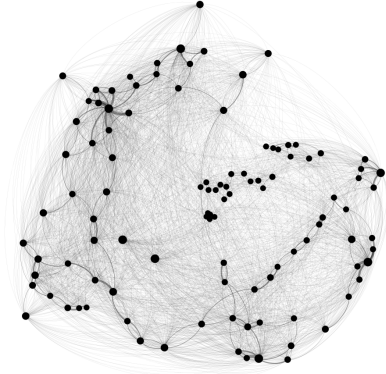


Figure 4: The Italy mobility network used in the simulations of Section 6. Each node is one of the $n = 107$ provinces in Italy, with edges representing mobility of people. Edge thickness is proportional to the edge weight.

Figs. 3a and 3b reveal that there are two locally exponentially stable boundary equilibria $(\bar{x}^1, \mathbf{0})$ or $(\mathbf{0}, \bar{x}^2)$, with convergence to either depending on the initial conditions. However, the DFE is unstable, and no trajectories in $\mathcal{D}$ converge to the DFE.

We present the next set of simulations, and then provide a brief summary of the key insights from all the simulations.

6.2. Real-world large-scale network

In this section, we present a simulation of the bivirus system with HOIs on a real-world large-scale network, illustrating a key phenomenon that arises precisely because there are HOIs present.

The network captures the mobility patterns of people traveling between provinces in Italy, shown in Fig. 4. Each of the $n = 107$ nodes represents a province in Italy, while the links capture the mobility and travel of people between provinces (including air, train, and road). The links are weighted and directed, representing the relative volume of travel between regions. The network is adapted from data of Parino et al. (2021), but with adjustments in order to fit the bivirus with HOI framework. We provide a brief summary below, with full details and code available at the following GitHub link.

Since the original network edge weights differ by several orders of magnitude, we remove all edges below a threshold value while ensuring the resulting network $\mathcal{G}$ is strongly

connected, with the adjacency matrix A being irreducible. We then set $A = A^1 = A^2$ and normalize the matrices so that the row sums are equal to 1. The two viruses are given different infection properties by setting heterogeneous values for β_1^k . In particular, we set $\beta_1^1 = 0.3$ and $\beta_1^2 = 0.15$. By setting $D^1 = D^2 = I$, we have that $\rho(\beta_1^k(D^k)^{-1}A^k) < 1$. Without HOIs, and as we discussed above, the DFE is globally exponentially stable and is the unique equilibrium in $\mathcal{D}$. However, as we shall see shortly, this no longer holds in the presence of HOIs.

To add HOIs into the network, we consider the undirected graph obtained from the square of $\mathcal{G}$, denoted by graph $\bar{\mathcal{G}}$. Let nodes i, j, k be a weakly connected triad (i, j, k) of $\bar{\mathcal{G}}$. We sample uniformly from the set of all triads of $\bar{\mathcal{G}}$, and for the sampled triad (i, j, k) , we add this as an HOI into the system by setting $b_{ijk} = 1$. A total of 300 HOIs were added to each of $\mathcal{G}^1$ and $\mathcal{G}^2$ (just under 10% of the pairwise edges), but because these were randomly sampled from the set of triads, the two viruses have different HOI graphs. We set $\beta_1^2 = 1.5$ and $\beta_2^2 = 2.5$, so the HOIs of the two viruses have different infection rates.

Figure 5 shows the simulation of the same network bivirus system with HOIs, under three different initial conditions; each initial condition involves sampling $x_i^k(0)$ from a uniform distribution of some interval $[a, b]$. The intervals are adjusted so that in Fig. 5a, $x_i^k(0) \in [0, 0.6]$ for $k = 1, 2$, in Fig. 5b, $x_i^1(0) \in [0, 0.8]$ and $x_i^2(0) \in [0, 0.1]$, and in Fig. 5c, $x_i^1(0) \in [0, 0.1]$ and $x_i^2(0) \in [0, 0.8]$. We can clearly see that the equilibrium to which the network converges can be different between the different initial conditions. The two single-virus endemic equilibria are both locally stable, but importantly, the DFE is also locally stable, as predicted by Proposition 2.

6.3. Remarks about the simulations

Several remarks about the simulations are in order. First, the majority of the propositions and theorems in Sections 4 and 5 are stated under Assumption 3, and yet in the simulations of Section 6.1, the same outcomes regarding stability and existence of various equilibria continue to hold under the more relaxed Assumption 2. This suggests that an important future direction is to formally prove the results in Sections 4 and 5 will hold under Assumption 2. Second, we have a number of example systems that demonstrate multi-stability, viz., having multiple attractive (locally exponentially stable) equilibria with different regions of attraction. In the classical bivirus systems *without* HOIs, multi-stable systems are more difficult to come by. They are often constructed by special design of the two-layer network to create strong asymmetry in the spread of virus 1 and virus 2 (Anderson and Ye, 2023; Ye et al., 2024). Here, all three examples appear to suggest that the presence of HOIs helps to facilitate a multi-stability property. This suggests a second line of future work, viz., further analysis of how HOIs can lead to novel dynamical phenomena in SIS models. Finally, of particular

note is the tristable equilibrium configuration with specifically the DFE being locally stable. See both Example 1 in Section 6.1 and also the real-world network in Section 6.2. Such a stability configuration is impossible in the classical bivirus system (without HOIs). Indeed, without HOIs, the network parameters in both simulation examples are such that the DFE is globally stable and there are no other equilibria; this is because the pairwise interactions are insufficiently intense, $\rho(\beta_1^k(D^k)^{-1}A^k) < 1$. The simulations thus highlight the fact that a single-virus endemic equilibrium can arise, and is in fact locally stable, if HOIs are sufficiently strong ($\beta_2^k \gg \beta_1^k$). Local stability suggests that one needs a sufficient concentration of the particular virus, in order for the HOIs to have a noticeable impact on the dynamics.

7. Conclusion

In this paper, we analyzed a competitive networked bivirus SIS HOI model. We obtained general results on equilibria and convergence under milder connectivity conditions compared to the literature, namely, for said general results, we do not require the pairwise interaction digraph to be strongly connected. We then derived new conditions for the stability and existence (or otherwise) of various equilibria. Several lines of research are worthy of further attention, as we have noted throughout the paper. We mention a few additional ones as well: First, the paper has not commented on conditions for global stability of single-virus endemic equilibria. Hence, one line of investigation may center around that. Second, provision of conditions for uniqueness and local stability of the coexistence equilibria discussed in Section 5 remains an open problem. Third, in the context of multi-virus HOI dynamics, the focus of the present paper has been on bivirus dynamics with the interactions being restricted to groups having size three. Hence, another thread of future work could focus on extending our modelling framework to account for arbitrary but finite number of competing viruses and/or interactions among groups having size n . Finally, devising control strategies for mitigating the spread of competing viruses with(out) accounting for HOI is another promising research direction.

References

- Anderson, B. D. and Ye, M. (2023), ‘Equilibria Analysis of a networked bivirus epidemic model using Poincaré–Hopf and Manifold Theory’, *SIAM Journal on Applied Dynamical Systems* **22**(4), 2856–2889.
- Aral, S. and Walker, D. (2011), ‘Creating social contagion through viral product design: A randomized trial of peer influence in networks’, *Management Science* **57**(9), 1623–1639.
- Ausiello, G. and Laura, L. (2017), ‘Directed hypergraphs: Introduction and fundamental algorithms—a survey’, *Theoretical Computer Science* **658**, 293–306.
- Battiston, F., Amico, E., Barrat, A., Bianconi, G., Ferraz de Arruda, G., Franceschiello, B., Iacopini, I., Kéfi, S., Latora, V., Moreno, Y. et al. (2021), ‘The physics of higher-order interactions in complex systems’, *Nature Physics* **17**(10), 1093–1098.

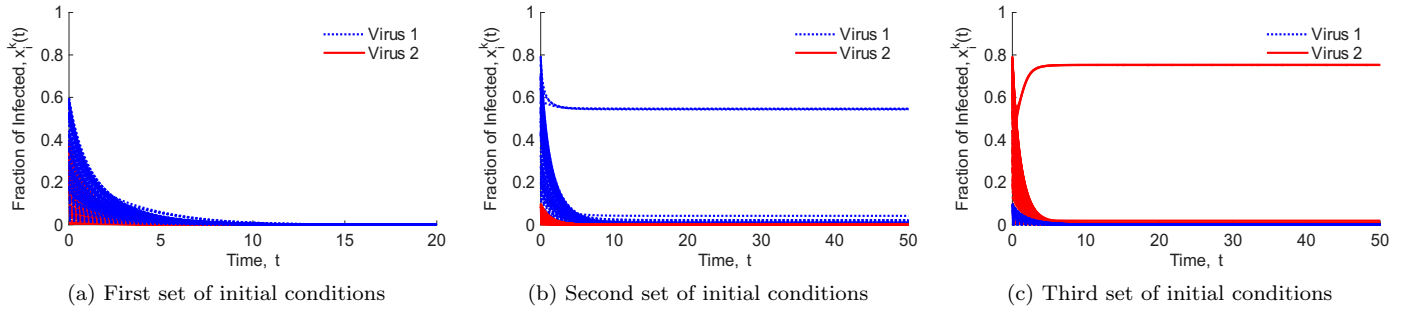


Figure 5: Trajectories of $x_i^k(t)$ for different simulations on the same network system. Blue dotted lines correspond to $x_i^1(t)$, while red solid lines correspond to $x_i^2(t)$. The panels correspond to three different sets of initial conditions, that result in convergence to different equilibria.

Battiston, F., Cencetti, G., Iacopini, I., Latora, V., Lucas, M., Patania, A., Young, J.-G. and Petri, G. (2020), ‘Networks beyond pairwise interactions: Structure and dynamics’, *Physics reports* **874**, 1–92.

Bonacich, P., Holdren, A. C. and Johnston, M. (2004), ‘Hyper-edges and multidimensional centrality’, *Social networks* **26**(3), 189–203.

Brauer, F., Van den Driessche, P. and Wu, J., eds (2008), *Mathematical Epidemiology*, Vol. 1945, Springer.

Bullo, F. (2024), *Lectures on Network Systems*, 1.7 edn, Kindle Direct Publishing.
URL: <https://fbullo.github.io/lns>

Castillo-Chavez, C., Huang, W. and Li, J. (1996), ‘Competitive exclusion in gonorrhea models and other sexually transmitted diseases’, *SIAM Journal on Applied Mathematics* **56**(2), 494–508.

Cisneros-Velarde, P. and Bullo, F. (2021), ‘Multigroup SIS epidemics with simplicial and higher order interactions’, *IEEE Transactions on Control of Network Systems* **9**(2), 695–705.

Cui, S., Liu, F., Liang, L., Jardón-Kojakhmetov, H. and Cao, M. (2025), ‘An sis diffusion process with direct and indirect spreading on a hypergraph’, *Automatica* **177**, 112319.

Cui, S., Zhang, G., Jardón-Kojakhmetov, H. and Cao, M. (2024), ‘On metzler positive systems on hypergraphs’, *arXiv preprint arXiv:2401.03652*.

de Arruda, G. F., Petri, G. and Moreno, Y. (2020), ‘Social contagion models on hypergraphs’, *Physical Review Research* **2**(2), 023032.

Doshi, V., Mallick, S. et al. (2022), ‘Convergence of Bi-Virus Epidemic Models With Non-Linear Rates on Networks – A Monotone Dynamical Systems Approach’, *IEEE/ACM Transactions on Networking*.

Ferguson, N., Laydon, D., Nedjati-Gilani, G., Imai, N., Ainslie, K., Baguelin, M., Bhatia, S., Boonyasiri, A., Cucunubá, Z., Cuomo-Dannenburg, G. et al. (2020), Report 9: Impact of non-pharmaceutical interventions (NPIs) to reduce COVID-19 mortality and healthcare demand, Technical Report 77482, Imperial College London.

Gibbs, T., Levin, S. A. and Levine, J. M. (2022), ‘Coexistence in diverse communities with higher-order interactions’, *Proceedings of the National Academy of Sciences* **119**(43), e2205063119.

Giordano, G., Blanchini, F., Bruno, R., Colaneri, P., Filippo, A. D., Matteo, A. D. and Colaneri, M. (2020), ‘Modelling the COVID-19 epidemic and implementation of population-wide interventions in Italy’, *Nature Medicine* **26**, 855–860.

Gracy, S., Anderson, B. D., Ye, M. and Uribe, C. A. (2024), Competitive networked bivirus SIS spread over hypergraphs, in ‘2024 American Control Conference (ACC)’, IEEE, pp. 4409–4415.

Gracy, S., Paré, P. E., Liu, J., Sandberg, H., Beck, C. L., Johansson, K. H. and Başar, T. (2025), ‘Modeling and analysis of a coupled SIS bi-virus model’, *Automatica* **171**, 111937.

Grilli, J., Barabás, G., Michalska-Smith, M. J. and Allesina, S. (2017), ‘Higher-order interactions stabilize dynamics in competitive network models’, *Nature* **548**(7666), 210–213.

Guillemin, V. and Pollack, A. (2010), *Differential Topology*, Vol. 370, American Mathematical Soc.

Gumel, A. B. and Song, B. (2008), ‘Existence of multiple-stable equilibria for a multi-drug-resistant model of mycobacterium tuberculosis’, *Mathematical Biosciences & Engineering*.

Hirsch, M. W. (1985), ‘Systems of differential equations that are competitive or cooperative ii: Convergence almost everywhere’, *SIAM Journal on Mathematical Analysis* **16**(3), 423–439.

Hirsch, M. W. (1988), ‘Stability and convergence in strongly monotone dynamical systems’, *Journal fur die reine und angewandte Mathematik*.

Hofbauer, J. (1990), ‘An index theorem for dissipative semiflows’, *The Rocky Mountain Journal of Mathematics* **20**(4), 1017–1031.

Iacopini, I., Petri, G., Barrat, A. and Latora, V. (2019), ‘Simplicial models of social contagion’, *Nature Communications* **10**(1), 2485.

Janson, A., Gracy, S., Paré, P. E., Sandberg, H. and Johansson, K. H. (2020), ‘Networked multi-virus spread with a shared resource: Analysis and mitigation strategies’, *arXiv:2011.07569*.

Khalil, H. (2002), *Nonlinear Systems*, Prentice Hall.

Klamt, S., Haus, U.-U. and Theis, F. (2009), ‘Hypergraphs and cellular networks’, *PLoS computational biology* **5**(5), e1000385.

Lajmanovich, A. and Yorke, J. A. (1976), ‘A deterministic model for gonorrhea in a nonhomogeneous population’, *Mathematical Biosciences* **28**(3-4), 221–236.

Lee, J. M. (2013), *Introduction to Smooth Manifolds*, Springer.

Lemieux, J. E., Siddle, K. J., Shaw, B. M., Loreth, C., Schaffner, S. F., Gladden-Young, A., Adams, G., Fink, T., Tomkins-Tinch, C. H., Krasilnikova, L. A. et al. (2021), ‘Phylogenetic analysis of sars-cov-2 in boston highlights the impact of superspreading events’, *Science* **371**(6529), eabe3261.

Liu, J., Paré, P. E., Nedić, A., Tang, C. Y., Beck, C. L. and Başar, T. (2019), ‘Analysis and control of a continuous-time bi-virus model’, *IEEE Transactions on Automatic Control* **64**(12), 4891–4906.

Mayfield, M. M. and Stouffer, D. B. (2017), ‘Higher-order interactions capture unexplained complexity in diverse communities’, *Nature ecology & evolution* **1**(3), 0062.

Myers, S. A. and Leskovec, J. (2012), Clash of the contagions: Cooperation and competition in information diffusion, in ‘Proceedings of the 12th International Conference on Data Mining’, IEEE, pp. 539–548.

Nickbakhsh, S., Mair, C., Matthews, L., Reeve, R., Johnson, P. C., Thorburn, F., Von Wissmann, B., Reynolds, A., McMenamin, J., Gunson, R. N. et al. (2019), ‘Virus–virus interactions impact the population dynamics of influenza and the common cold’, *Proceedings of the National Academy of Sciences* **116**(52), 27142–27150.

Nuno, M., Feng, Z., Martcheva, M. and Castillo-Chavez, C. (2005), ‘Dynamics of two-strain influenza with isolation and partial cross-immunity’, *SIAM Journal on Applied Mathematics* **65**(3), 964–982.

Paré, P. E., Liu, J., Beck, C. L., Nedić, A. and Başar, T. (2021), ‘Multi-competitive viruses over time-varying networks with mutations and human awareness’, *Automatica* **123**, 109330.

Parino, F., Zino, L., Porfiri, M. and Rizzo, A. (2021), ‘Modelling and predicting the effect of social distancing and travel restrictions on COVID-19 spreading’, *Journal of the Royal Society Interface*

18(175), 20200875.

- Pastor-Satorras, R., Castellano, C., Van Mieghem, P. and Vespignani, A. (2015), ‘Epidemic processes in complex networks’, *Reviews of modern physics* **87**(3), 925.
- Rantzer, A. (2011), Distributed control of positive systems, in ‘Proceedings of the 50th IEEE Conference on Decision and Control’, pp. 6608–6611.
- Sahneh, F. D. and Scoglio, C. (2014), ‘Competitive epidemic spreading over arbitrary multilayer networks’, *Physical Review E* **89**(6), 062817.
- Shapiro, J. H. and Shapiro, J. H. (2016), ‘Nash equilibrium: Fixed points in game theory’, *A Fixed-Point Farrago* pp. 51–64.
- Smith, H. L. (1986), ‘Competing subcommunities of mutualists and a generalized kamke theorem’, *SIAM Journal on Applied Mathematics* **46**(5), 856–874.
- Smith, H. L. (1988), ‘Systems of ordinary differential equations which generate an order preserving flow. a survey of results’, *SIAM Review* **30**(1), 87–113.
- Stella, L., Martínez, A. P., Bauso, D. and Colaneri, P. (2022), ‘The role of asymptomatic infections in the covid-19 epidemic via complex networks and stability analysis’, *SIAM Journal on Control and Optimization* **60**(2), S119–S144.
- Trpevski, D., Tang, W. K. and Kocarev, L. (2010), ‘Model for rumor spreading over networks’, *Physical Review E* **81**(5), 056102.
- Ugander, J., Backstrom, L., Marlow, C. and Kleinberg, J. (2012), ‘Structural diversity in social contagion’, *Proceedings of the national academy of sciences* **109**(16), 5962–5966.
- Varga, R. (1999), *Matrix Iterative Analysis*, Springer Series in Computational Mathematics, Springer Berlin Heidelberg.
- Wang, W., Nie, Y., Li, W., Lin, T., Shang, M.-S., Su, S., Tang, Y., Zhang, Y.-C. and Sun, G.-Q. (2024), ‘Epidemic spreading on higher-order networks’, *Physics Reports* **1056**, 1–70.
- Ye, M. and Anderson, B. D. (2022), ‘Competitive epidemic spreading over networks’, *IEEE Control Systems Letters* **7**, 545–552.
- Ye, M., Anderson, B. D., Janson, A., Gracy, S. and Johansson, K. H. (2024), ‘Competitive epidemic networks with multiple survival-of-the-fittest outcomes’, *Systems & Control Letters* **193**, 105907.
- Ye, M., Anderson, B. D. and Liu, J. (2022), ‘Convergence and equilibria analysis of a networked bivirus epidemic model’, *SIAM Journal on Control and Optimization* **60**(2), S323–S346.
- Zino, L. and Rizzo, A. (2024), On a susceptible-infected-susceptible epidemic model with reactive behavioral response on higher-order temporal networks, in ‘2024 IEEE 63rd Conference on Decision and Control (CDC)’, IEEE, pp. 3912–3917.

Appendix: Proof of Theorem 2

We treat first the case where the A^k and B^k are fixed. Consider (2) in the shorthand form $\dot{x} = f(x)$, and to emphasise the possibility of variations in the entries of D^k , show this explicitly by writing the equations as $\dot{x} = f_\delta(x, \delta)$, where δ is a vector containing all the diagonal entries of the D^k . It is easily established that

$$\frac{\partial f_\delta(x^1, x^2, D^1, D^2)}{\partial \delta} = \frac{\partial f_\delta(x^1, x^2, D^1, D^2)}{\partial (D^1, D^2)} = \begin{bmatrix} X^1 & 0 \\ 0 & X^2 \end{bmatrix} \quad (40)$$

At a coexistence equilibrium, from Lemma 2, it is clear that this matrix has full row rank. Consequently, the matrix $\frac{\partial f_\delta(x, \delta)}{\partial (x, \delta)}$ (which is obtained simply by adding further columns) also has full row rank at a coexistence equilibrium.

Similarly, in preparation for considering the second claim of the theorem, we can consider f as depending explicitly on the possibly variable A^k and B^k . It is helpful

to introduce a rewriting of the key system equation (2). Define for $k = 1, 2$ the matrices

$$C^k(x^k) = \beta_1^k A^k + \beta_2^k \begin{bmatrix} (x^k)^\top B_1^k \\ (x^k)^\top B_2^k \\ \vdots \\ (x^k)^\top B_n^k \end{bmatrix} \quad (41)$$

This means that the system equations can be written as

$$\begin{aligned} \dot{x}^1 &= -D^1 x^1 + (I - X^1 - X^2) C^1(x^1) x^1 \\ \dot{x}^2 &= -D^2 x^2 + (I - X^1 - X^1) C^2(x^2) x^2 \end{aligned} \quad (42)$$

There are a finite number of patterns of zero entries for the A^k and B^k for which the hypergraph is strongly connected. When the hypergraph is strongly connected, by Corollary 2 for any positive x^k the matrices C^k are irreducible. For each matrix separately, arbitrarily choose any one such pattern, and let γ^k denote the vector of nonzero entries of the C^k ; also set $\gamma = [(\gamma^1)^\top (\gamma^2)^\top]^\top$. Denote the associated f as $f_\gamma(x, \gamma)$. Then, one can show that

$$\frac{\partial f_\gamma(x, \gamma)}{\partial \gamma} = \begin{bmatrix} I - X^1 - X^2 & 0 \\ 0 & I - X^1 - X^2 \end{bmatrix} \begin{bmatrix} \frac{\partial(C^1(x^1)x^1)}{\partial \gamma^1} & 0 \\ 0 & \frac{\partial(C^2(x^2)x^2)}{\partial \gamma^2} \end{bmatrix} \quad (43)$$

A detailed calculation as set out in (Anderson and Ye, 2023) shows that the second matrix in the product (on the right-hand side of (43)) has full row rank. (The irreducibility of each $C^k(x^k)$ for any positive x^k ensures in particular that every row is nonzero, and this is the key property which guarantees the conclusion.) We also know from Lemma 2 that no equilibrium of (2) can cause any diagonal entry of $I - X^1 - X^2$ to be zero, which means that the first matrix of the product is nonsingular. Hence, the matrix on the left has full row rank at any equilibrium. The matrix $\frac{\partial f_\gamma(x, \gamma)}{\partial (x, \gamma)}$, obtained by the addition of further columns to $\frac{\partial f_\gamma(x, \gamma)}{\partial \gamma}$, then has full row rank.

We now complete the proof for the first part of the theorem. (Proof of the second part is virtually identical). Without loss of generality, suppose that for some $\bar{\delta}$, all diagonal entries δ_i of the D^k satisfy $0 < \bar{\delta} < \delta_i < \bar{\delta}^{-1}$. Denote the set of such δ as $\mathcal{D}$. Call the set of x in the interior of the region of interest $\mathcal{X}$. Then $\mathcal{X} \times \mathcal{D}$ is a manifold, being a product of open sets, and $f_\delta(x, \delta)$ is a mapping of the manifold $\mathcal{X} \times \mathcal{D}$. Let the image be denoted by $\mathcal{Y}$, and let $\mathcal{Z}$ be the submanifold of $\mathcal{Y}$ consisting of the single point $\mathbf{0}_{2n}$. (The set $f_\delta^{-1}(\mathcal{Z})$ is the set of zeros of f .) Since the Jacobian of f_δ with respect to (x, δ) has full row rank on this set, this means that the map f_δ is transversal to $\mathcal{Z}$. By the Parametric Transversality Theorem, this means that for all choices of $\delta = \delta^*$ excluding a set of measure zero, the Jacobian $\frac{\partial f_{\delta^*}(x, \delta^*)}{\partial x}$ will be nonsingular at a preimage of $\mathcal{Z}$, i.e at a zero. This is equivalent to saying that the zeros are nondegenerate. The boundedness of the region $\mathcal{X}$ then guarantees that there can only be a finite number of such zeros.